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Selzer

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(54) **MULTIFUNCTIONAL DESIGN HOUSING COMPONENTS**

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(51) **Int. Cl.**

E04C 1/39	(2006.01)
E04C 3/04	(2006.01)
E04F 10/02	(2006.01)
E04F 17/08	(2006.01)

(52) **U.S. Cl.**

CPC ***E04C 1/39*** (2013.01); ***E04C 3/04*** (2013.01); ***E04F 10/02*** (2013.01); ***E04F 17/08*** (2013.01); *E04C 2003/0404* (2013.01)

(58) **Field of Classification Search**

CPC E04C 1/39; E04C 2003/0404; E04C 3/04; E04C 2003/0417; E04C 2003/043; E04C 2003/0465; E04C 3/28; E04F 10/02; E04F 17/08; E04F 2019/044; E04F 10/10

USPC 52/287.1, 288.1

See application file for complete search history.

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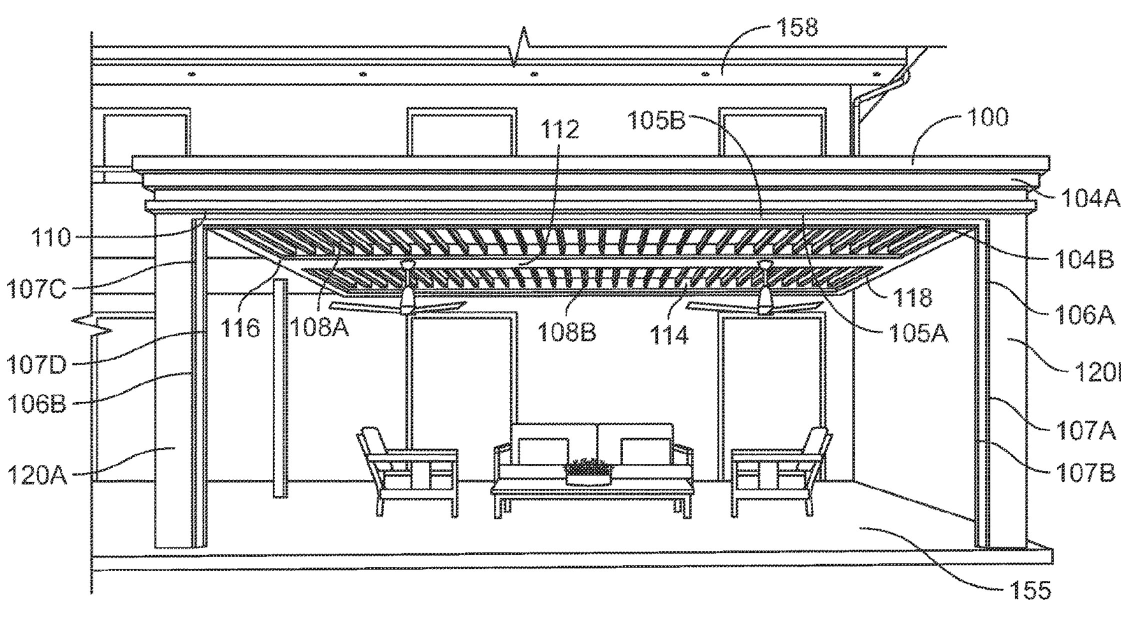
(57) **ABSTRACT**

A multifunctional design housing is disclosed. In some examples, the multifunctional design housing includes an outer case with a back wall, a first side wall, and a second side wall. The back wall includes a first back wall edge and a second back wall edge. The first side wall extends forward from the first back wall edge, and the second side wall extends forward from the second back wall edge. The front wall includes a first side wall front edge and a second side wall front edge. A front opening is formed between the first side wall front edge and the second side wall front edge providing access to an interior volume within the outer case. A conduit control arm with a support element is spaced forward from the back wall and coupled to the outer case by a spacing element.

19 Claims, 24 Drawing Sheets



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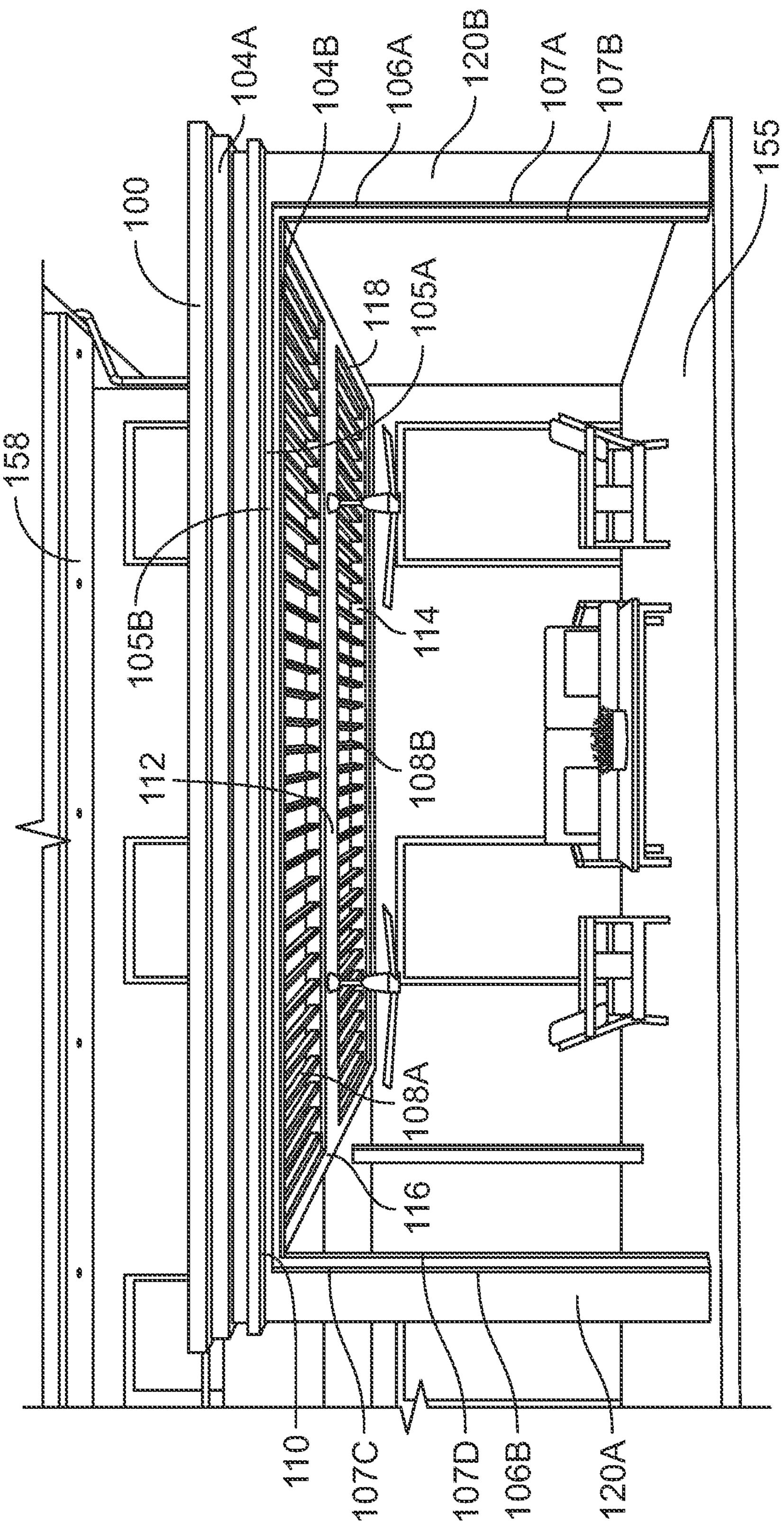
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107C
107D
106B
120A

FIG. 1

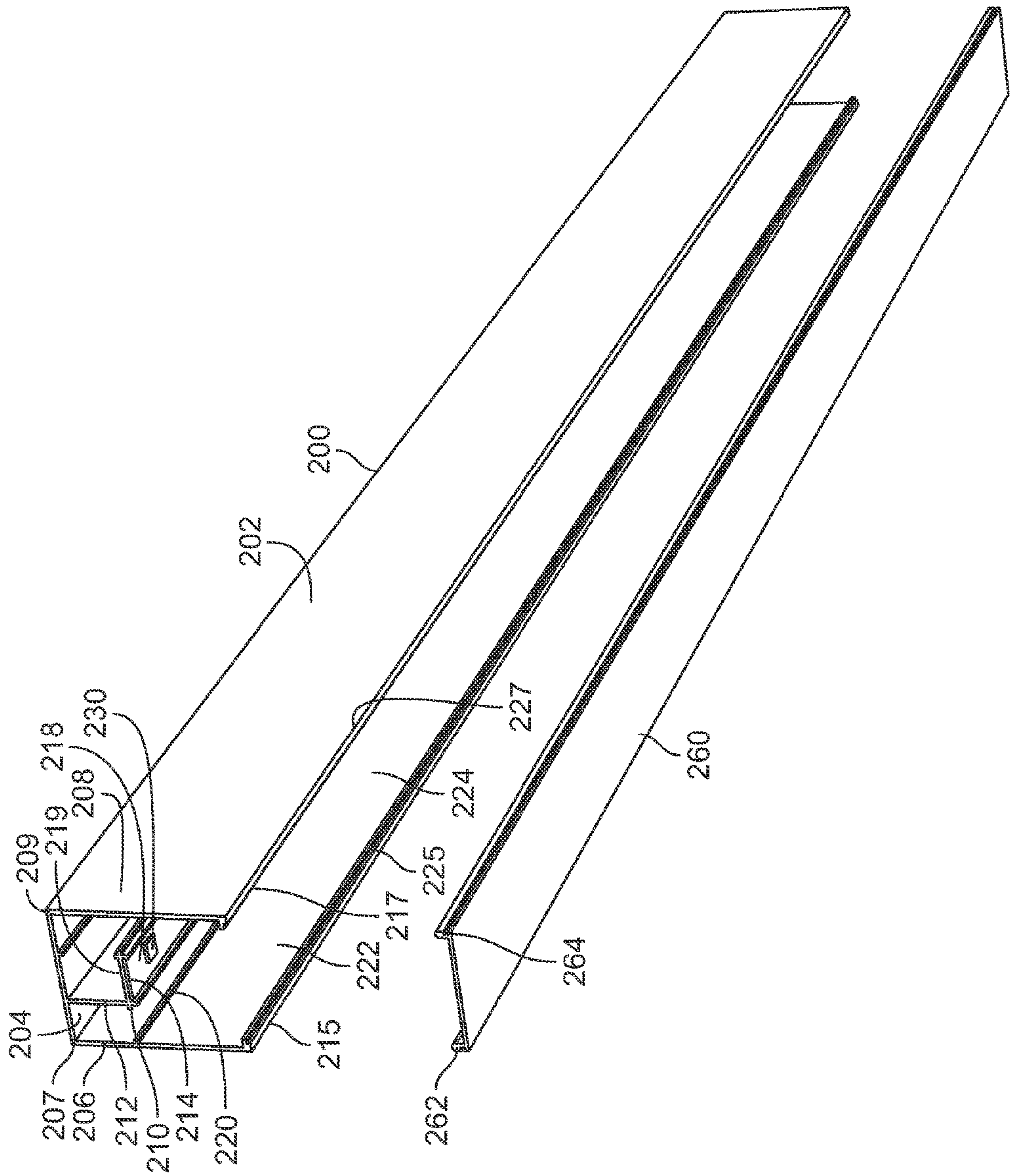
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FIG. 2A

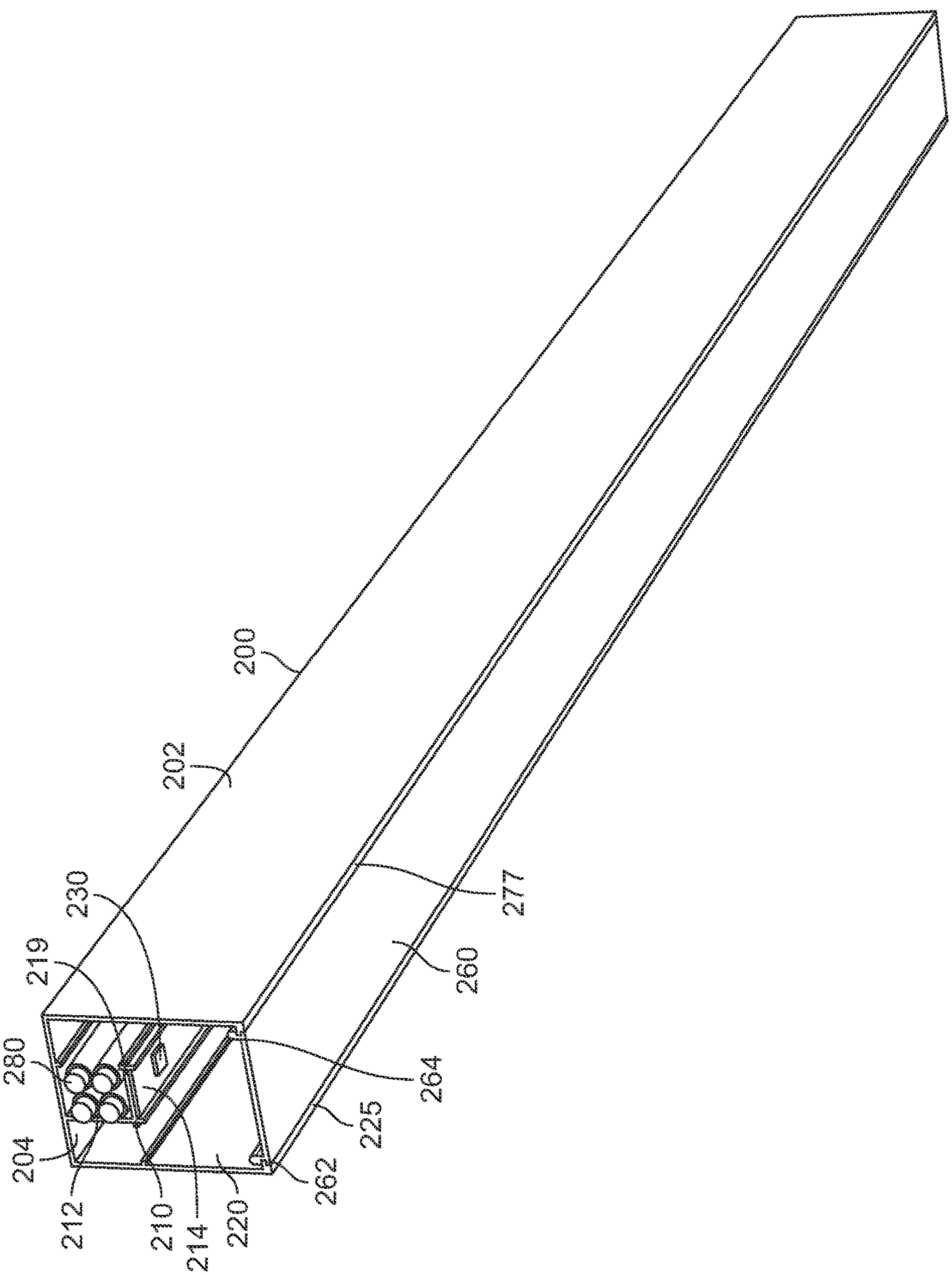

FIG. 2B

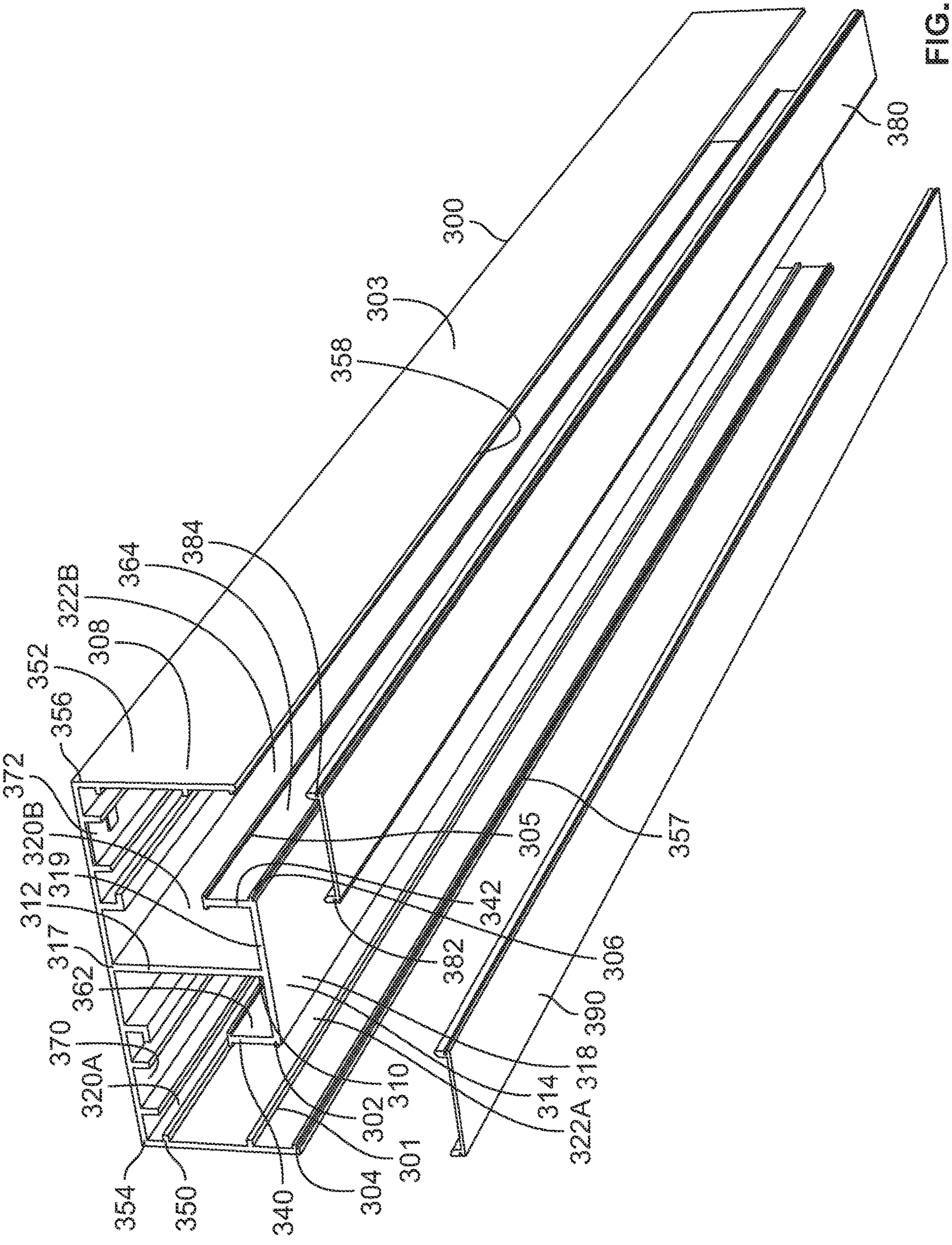
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FIG. 3A

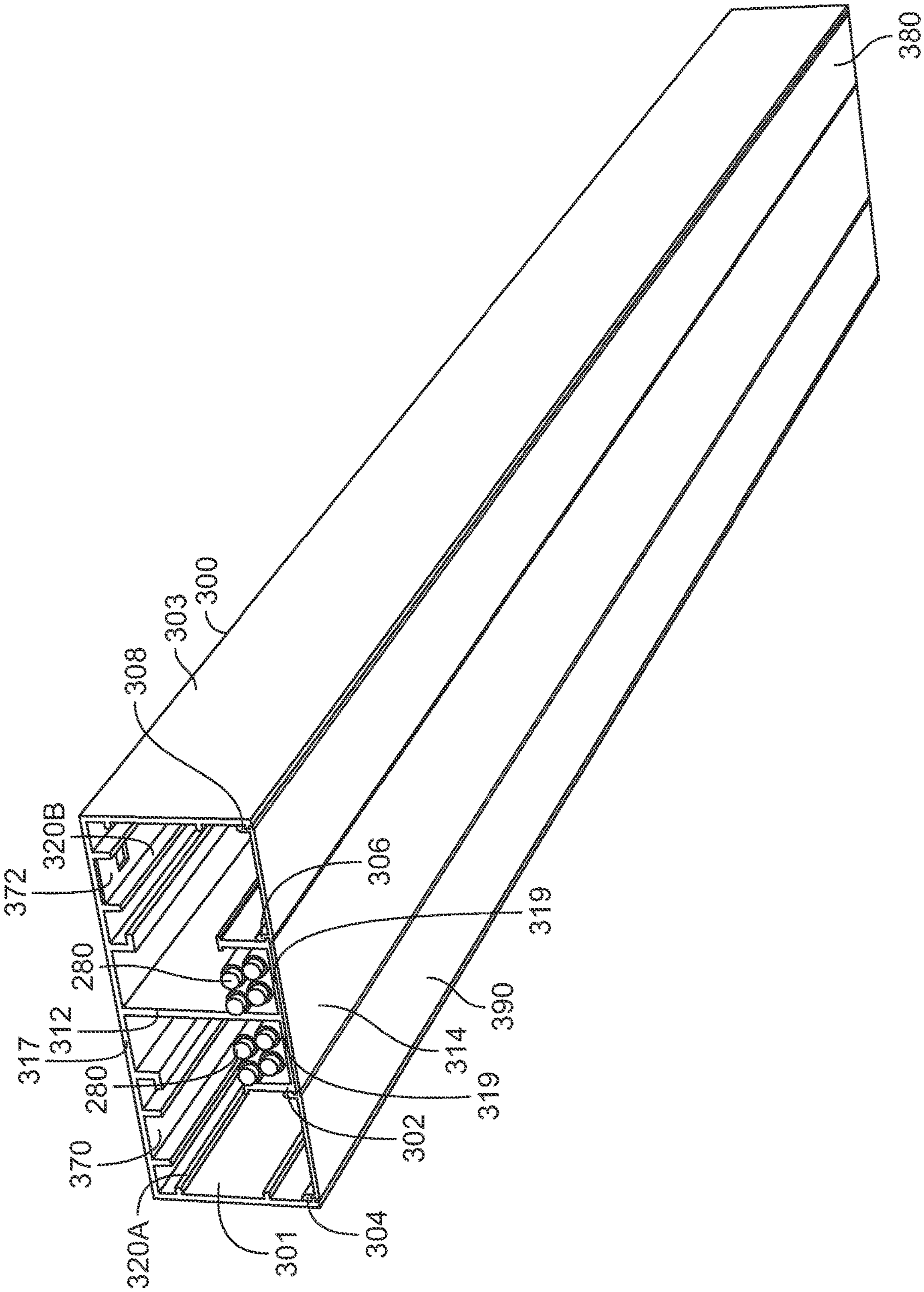
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FIG. 3B

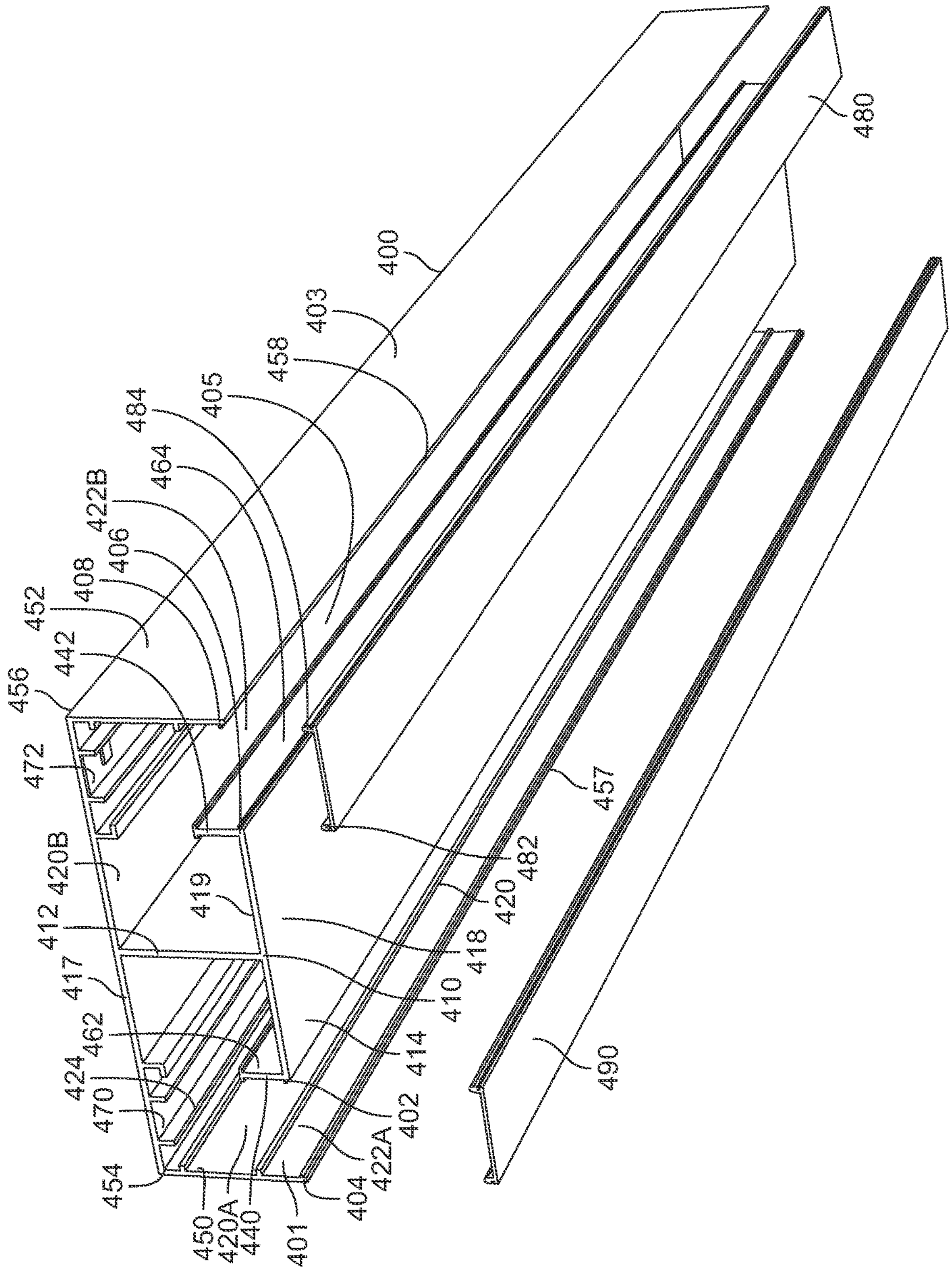

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FIG. 4A

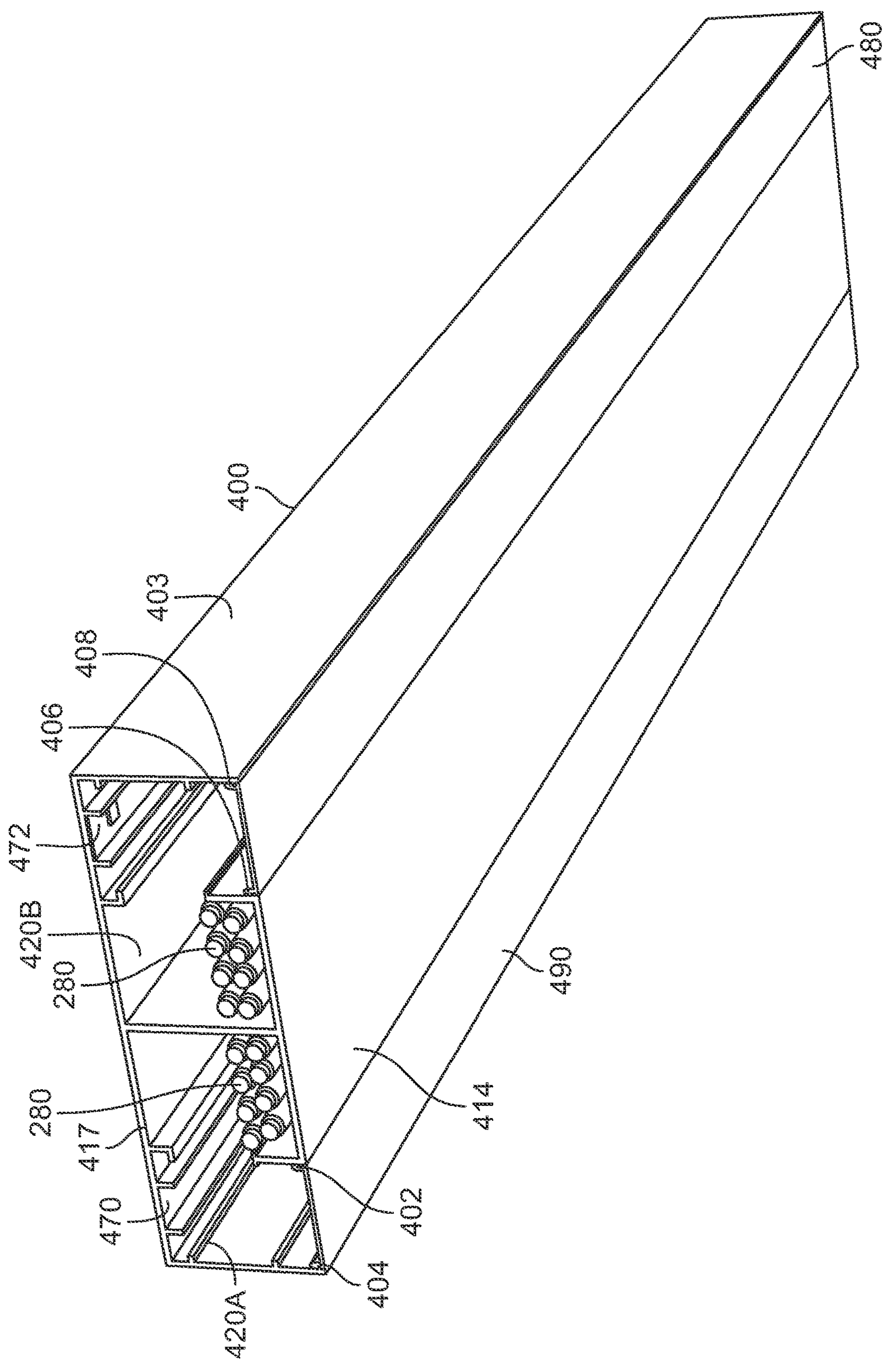
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FIG. 4B

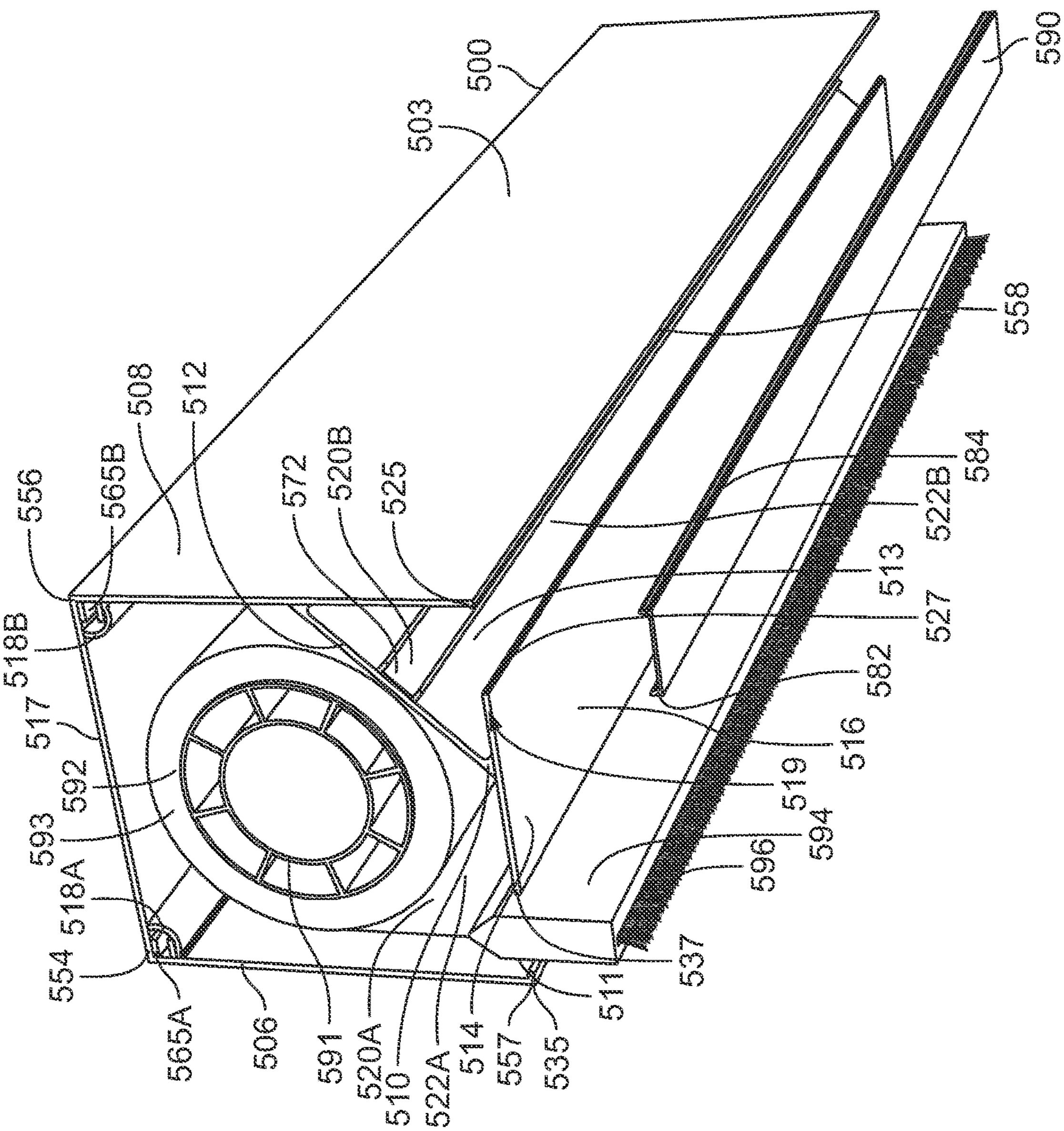

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FIG. 5A

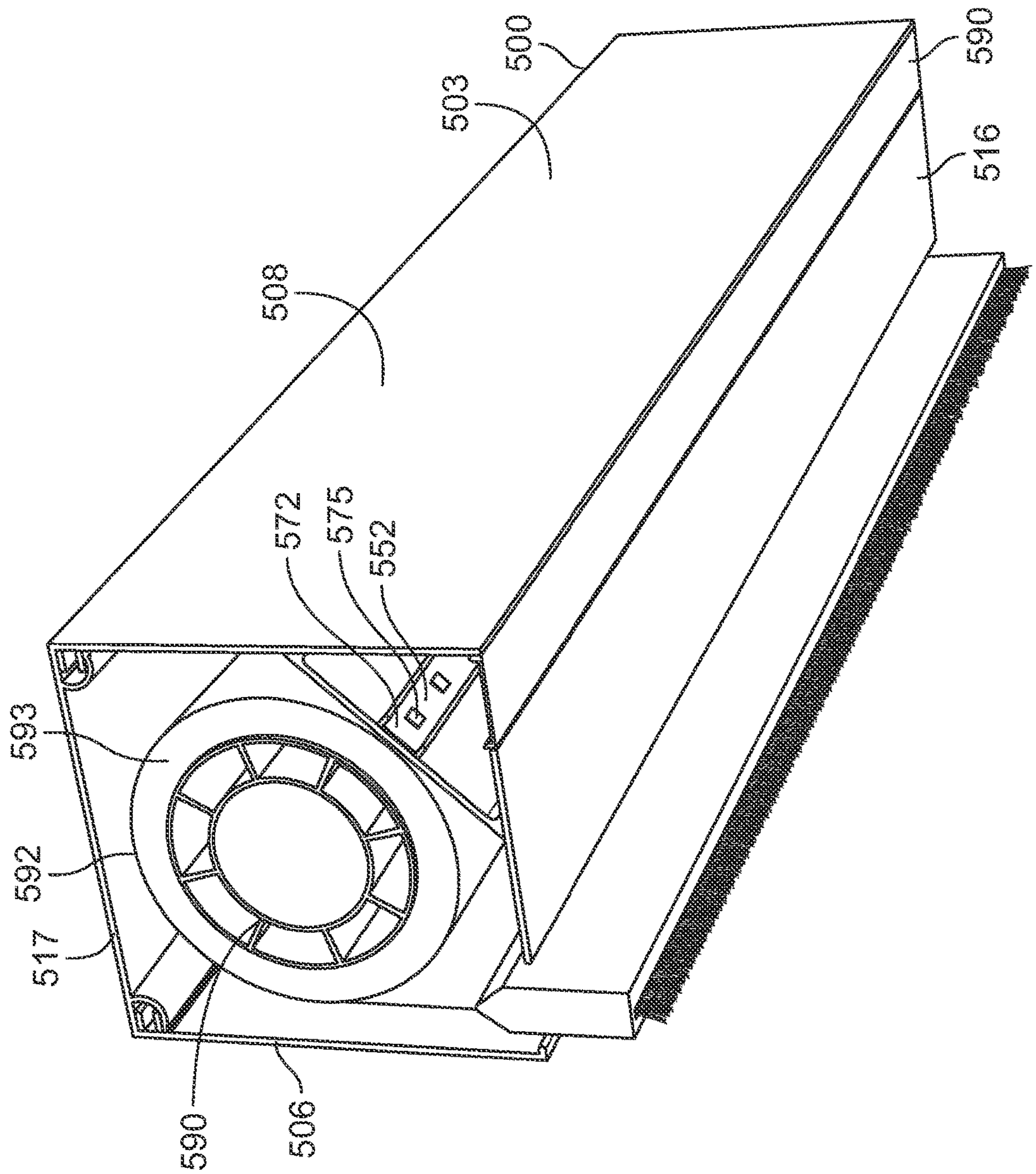

FIG. 5B

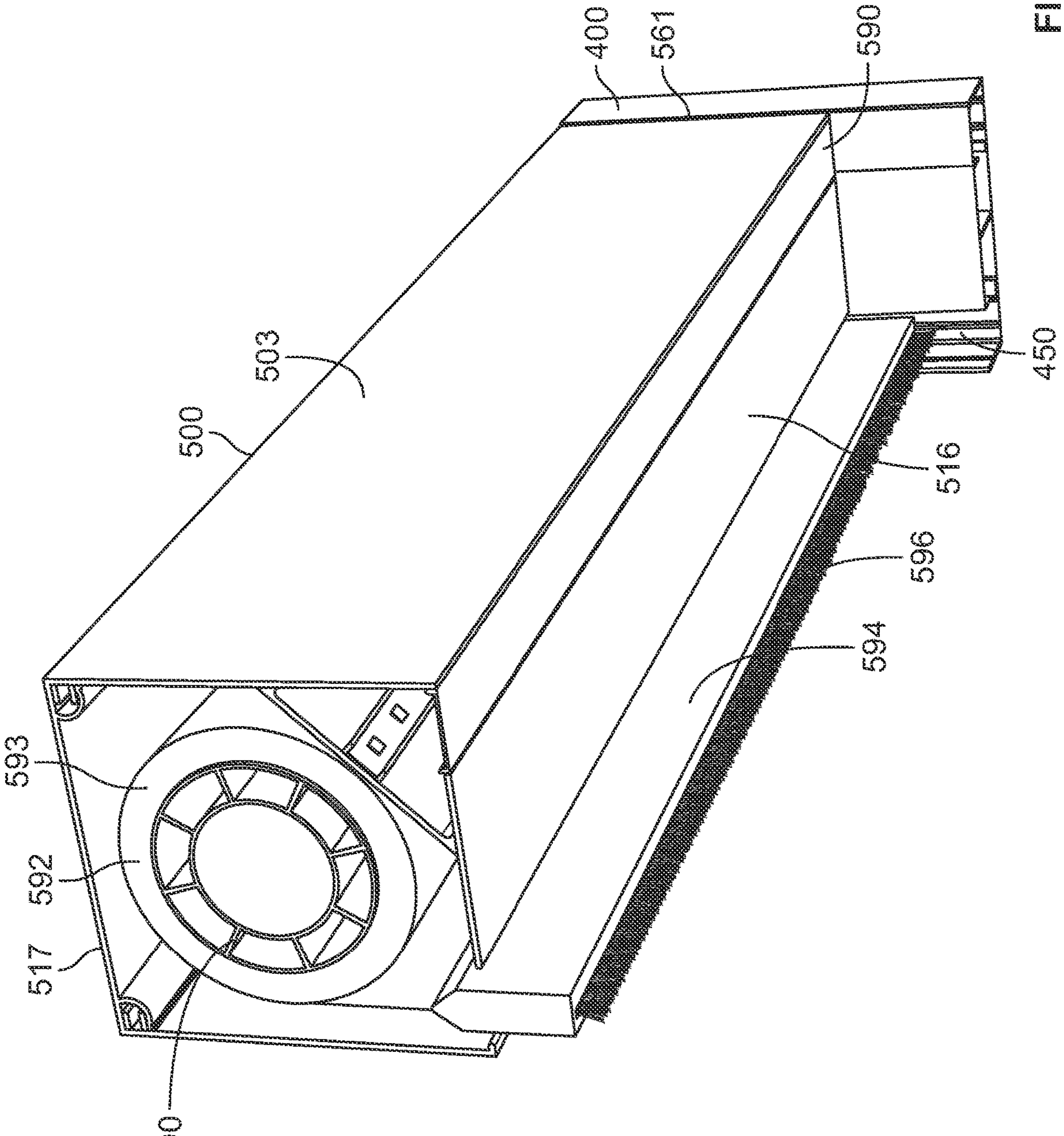

FIG. 5C

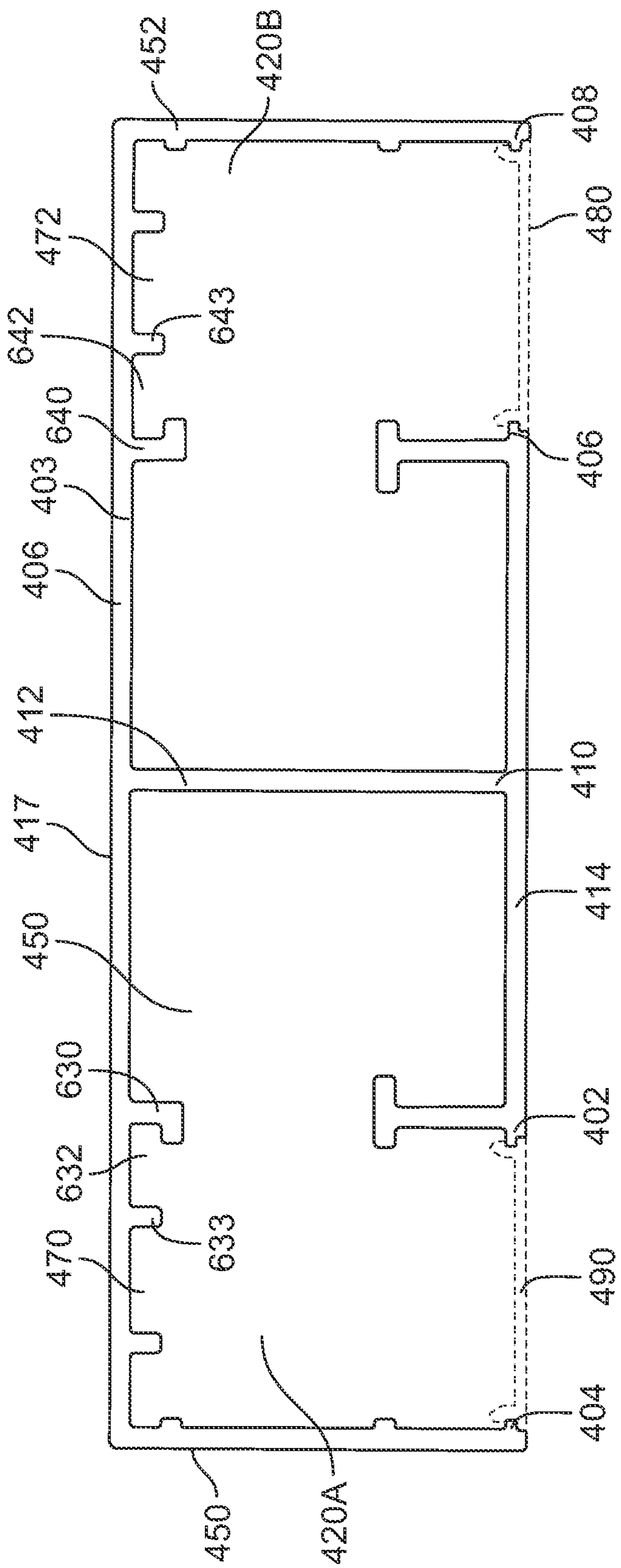
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FIG. 6A

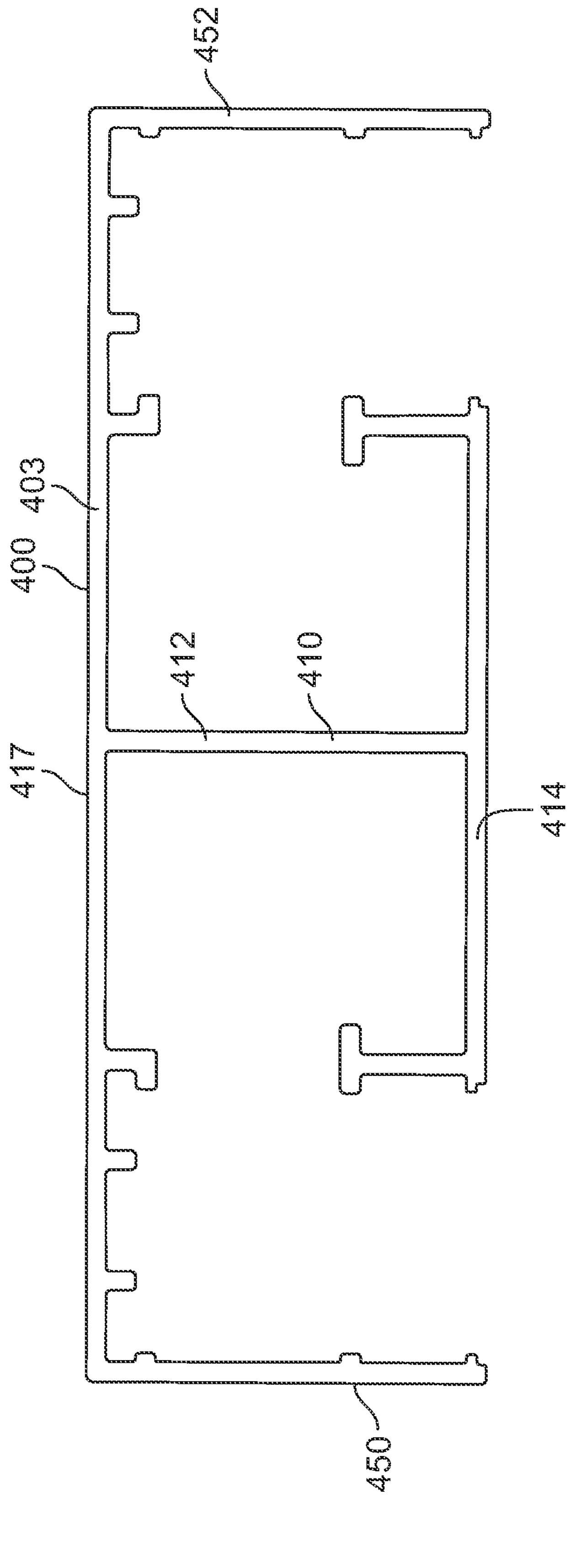

FIG. 6B

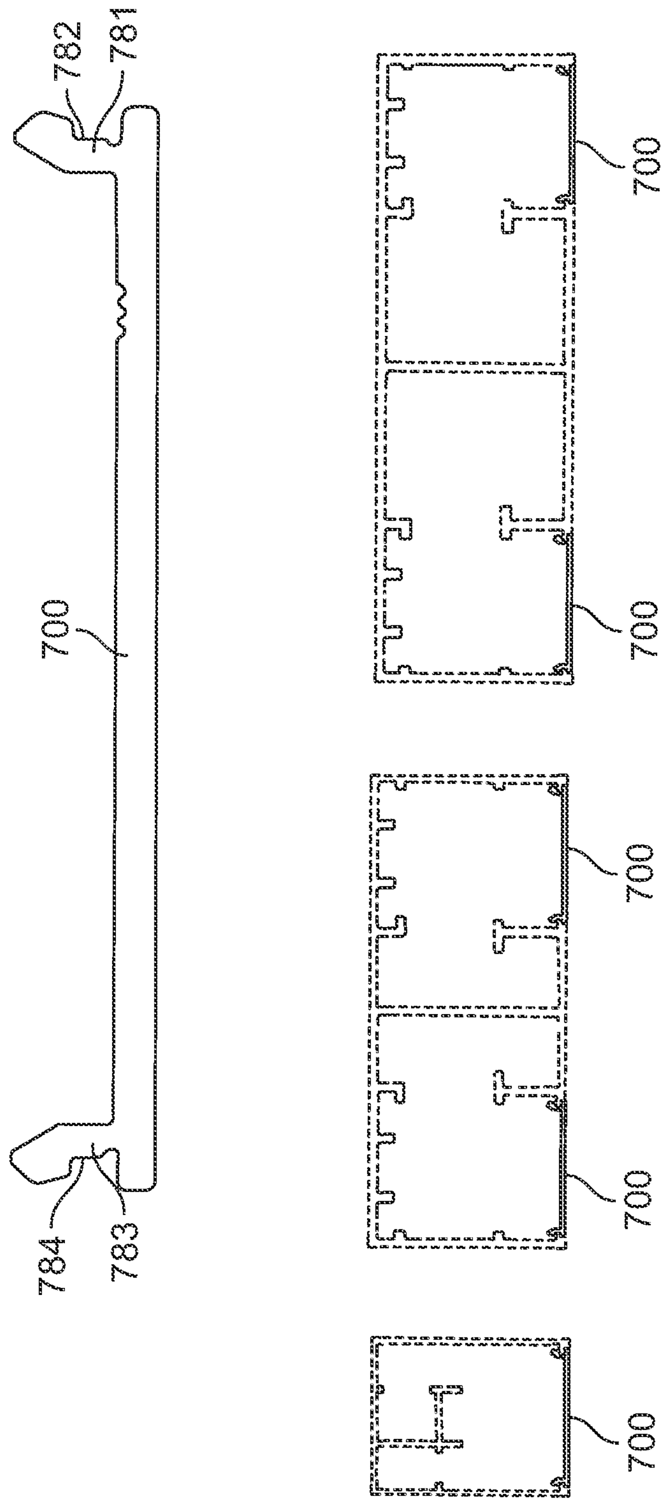

FIG. 7

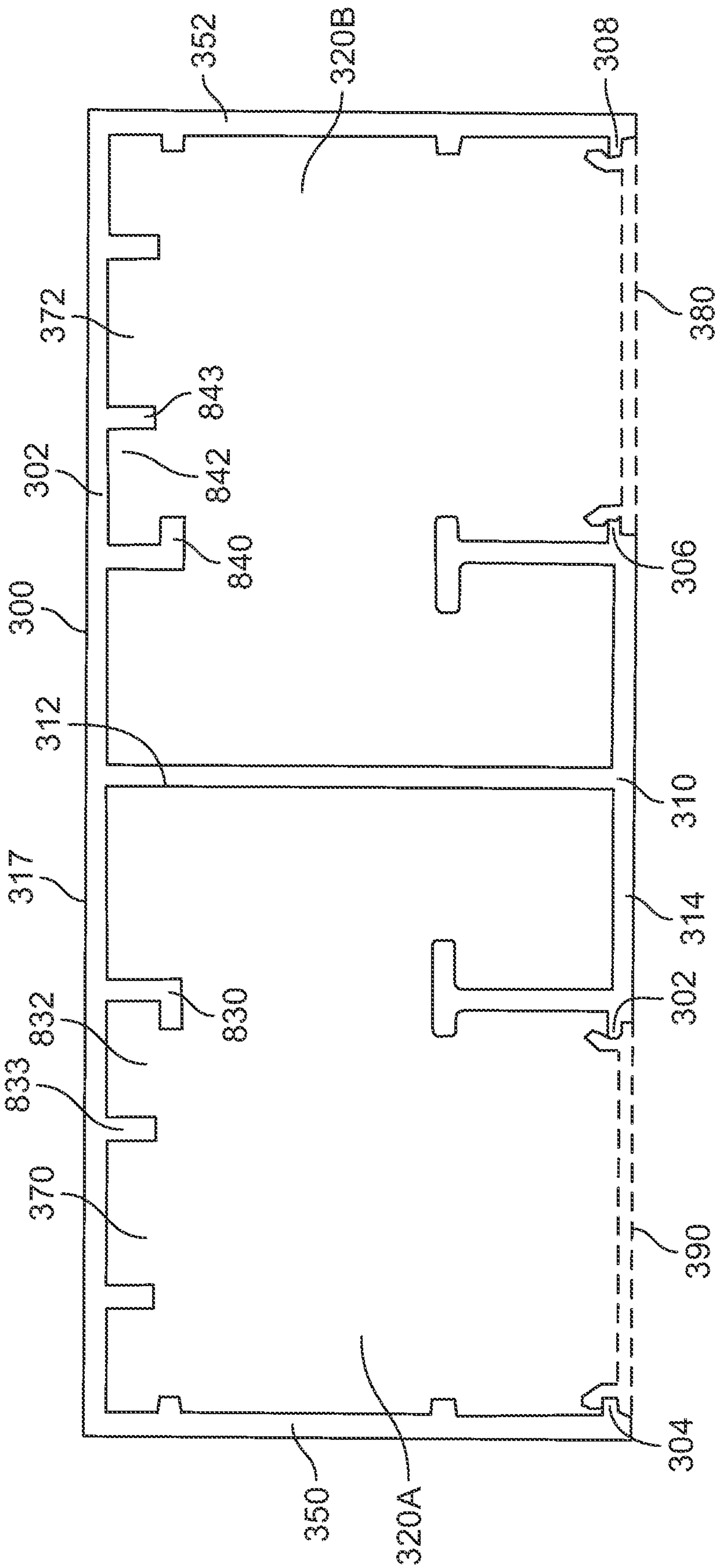
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FIG. 8A

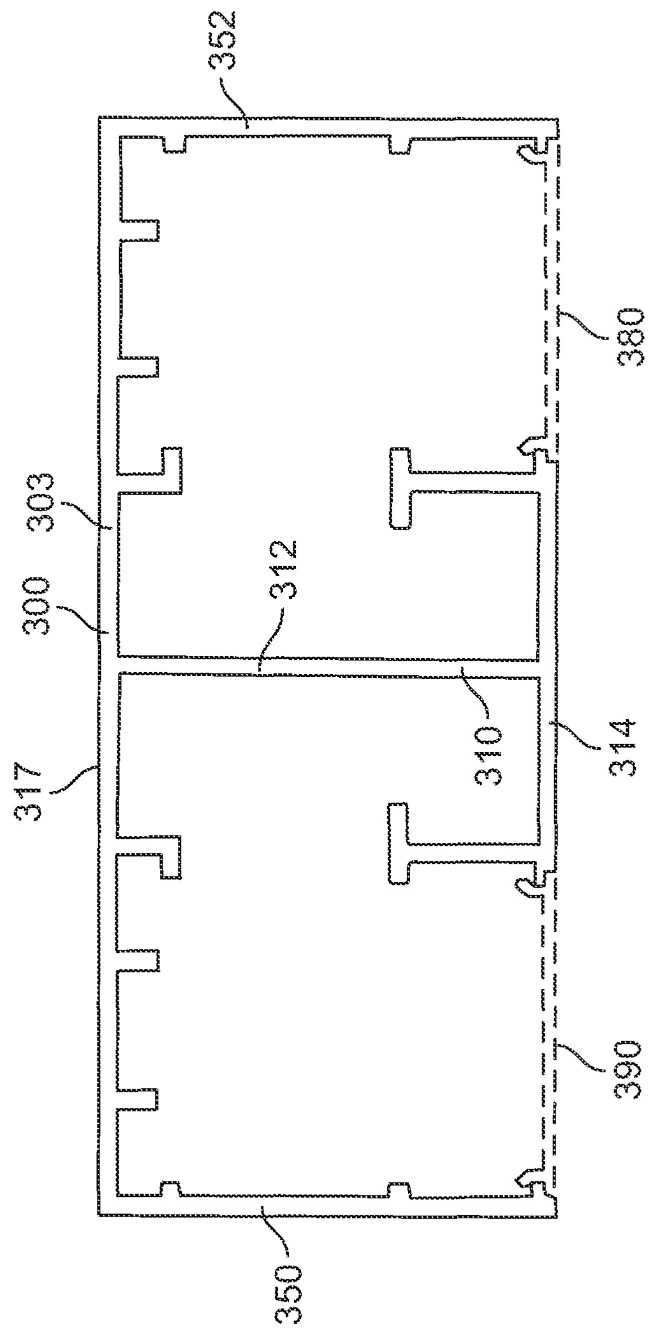

FIG. 8B

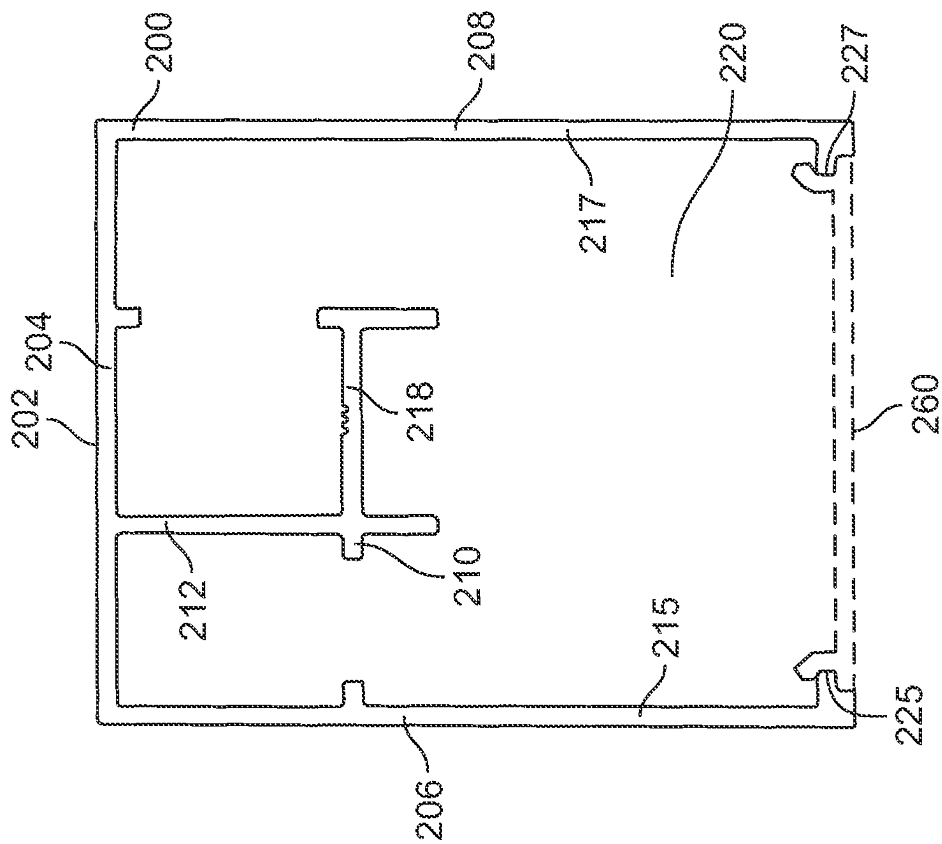

FIG. 9

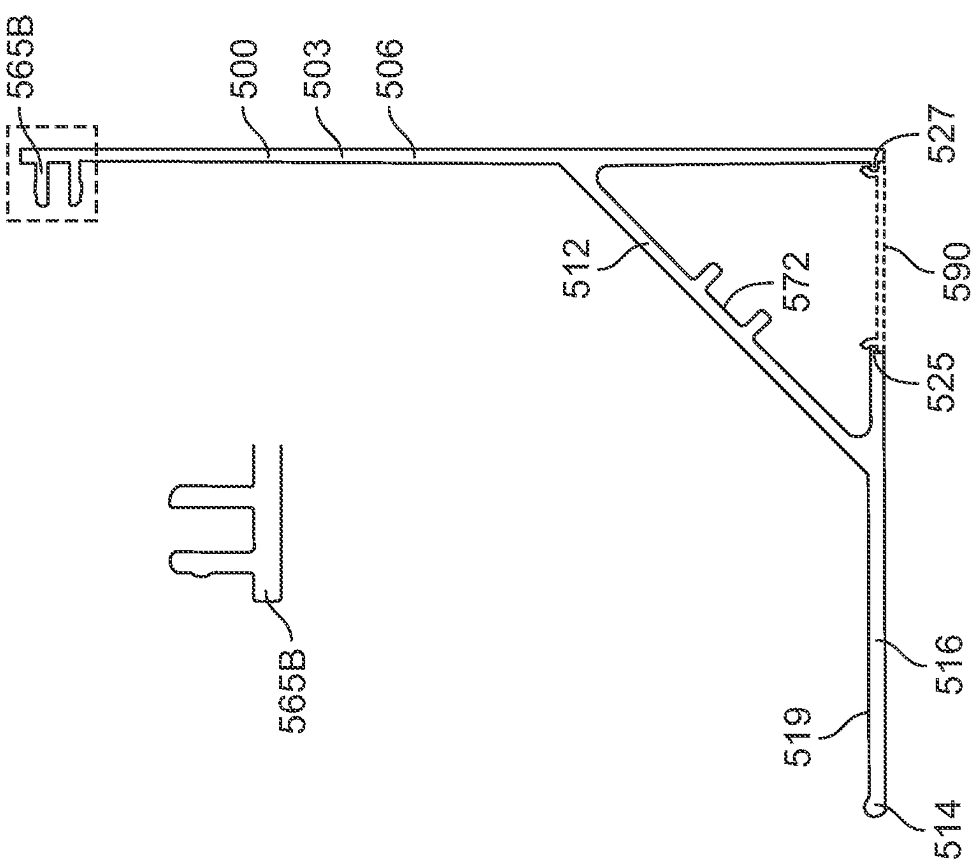

FIG. 10A

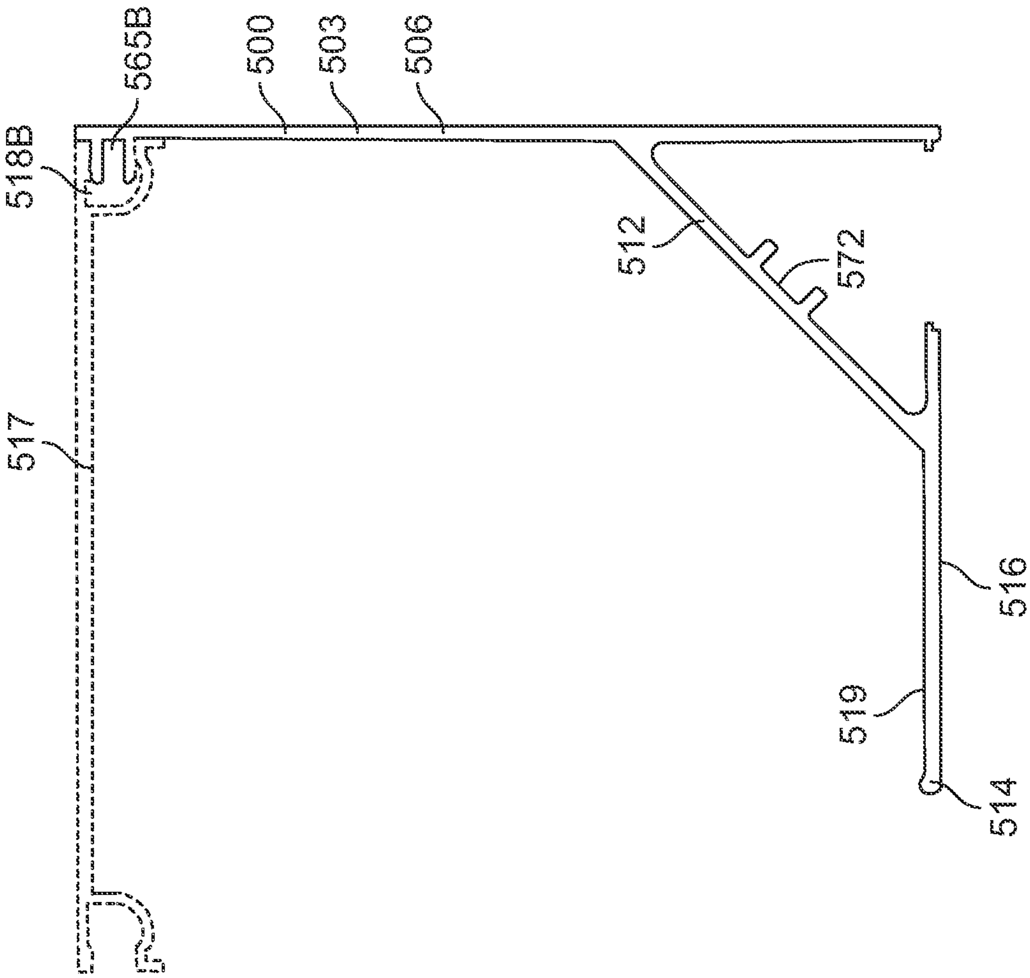

FIG. 10B

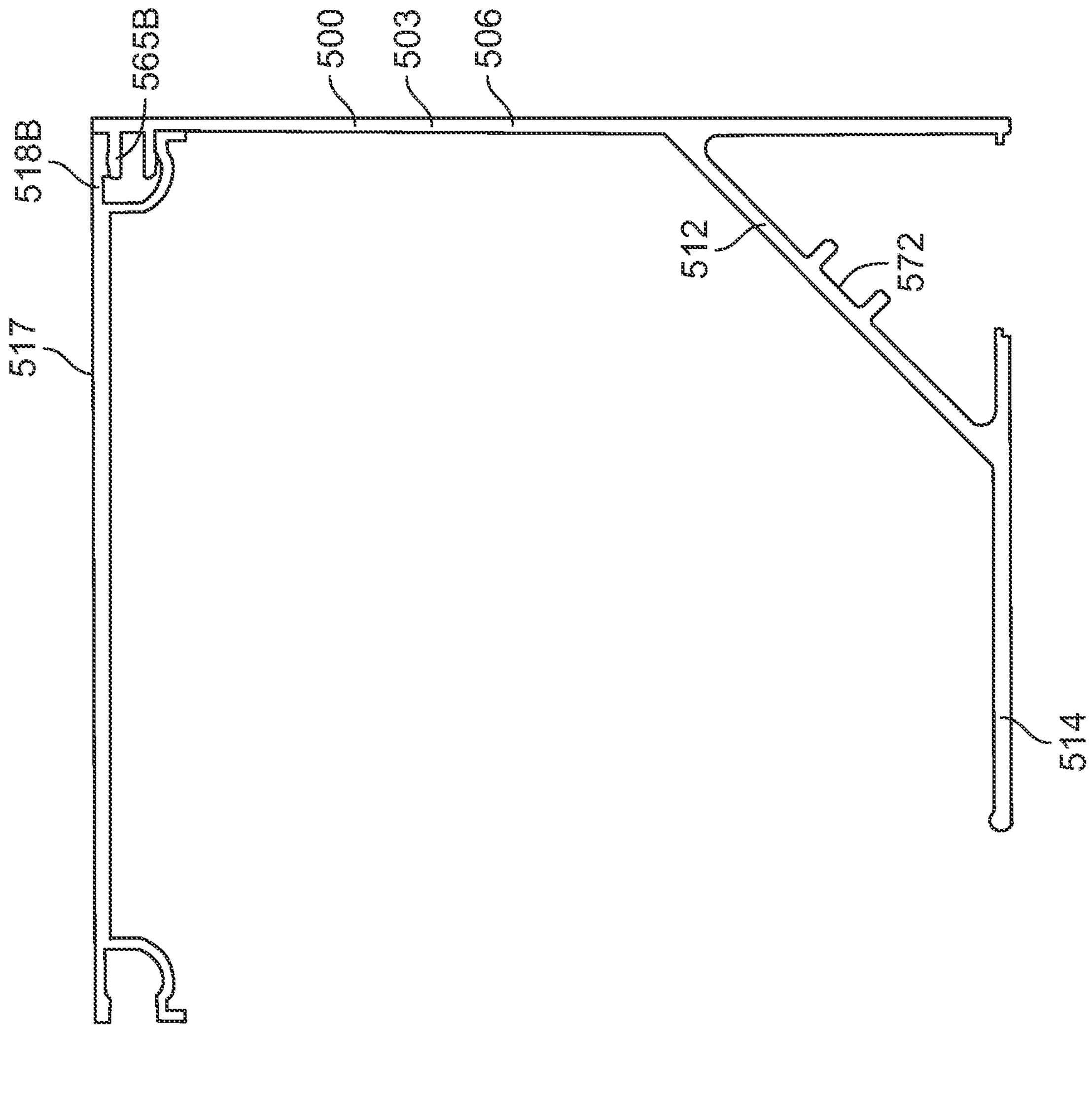

FIG. 10C

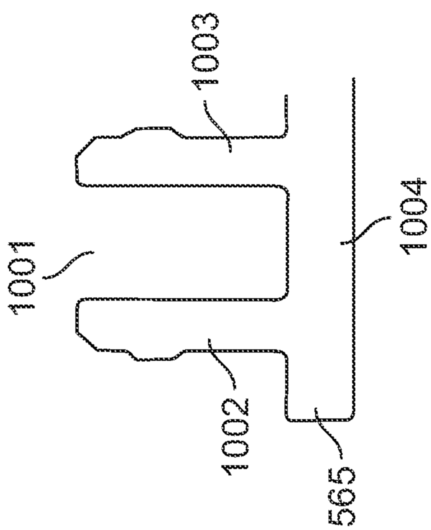

FIG. 10D

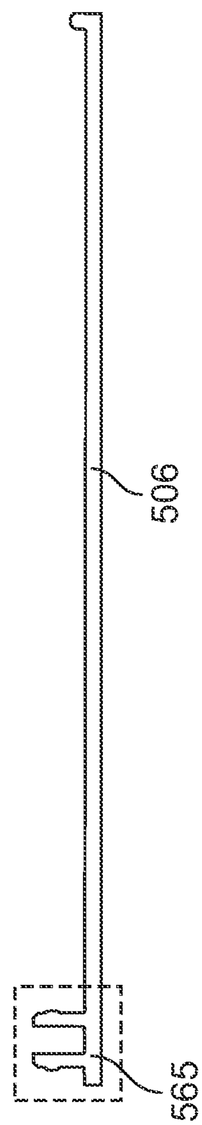

FIG. 10E

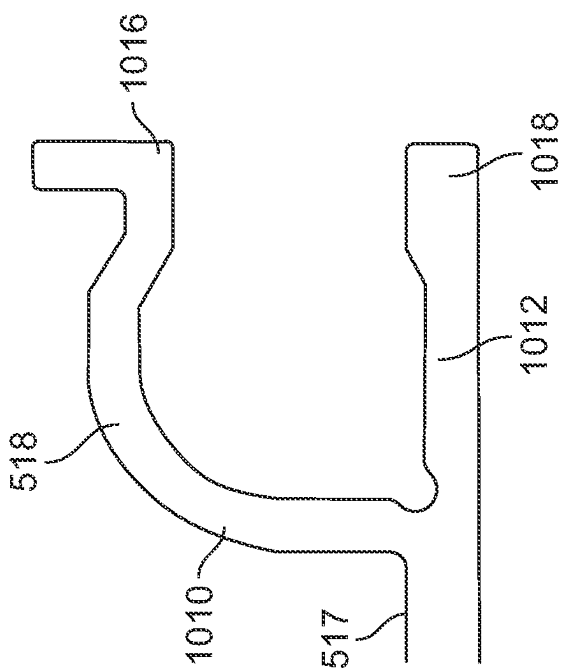

FIG. 10F

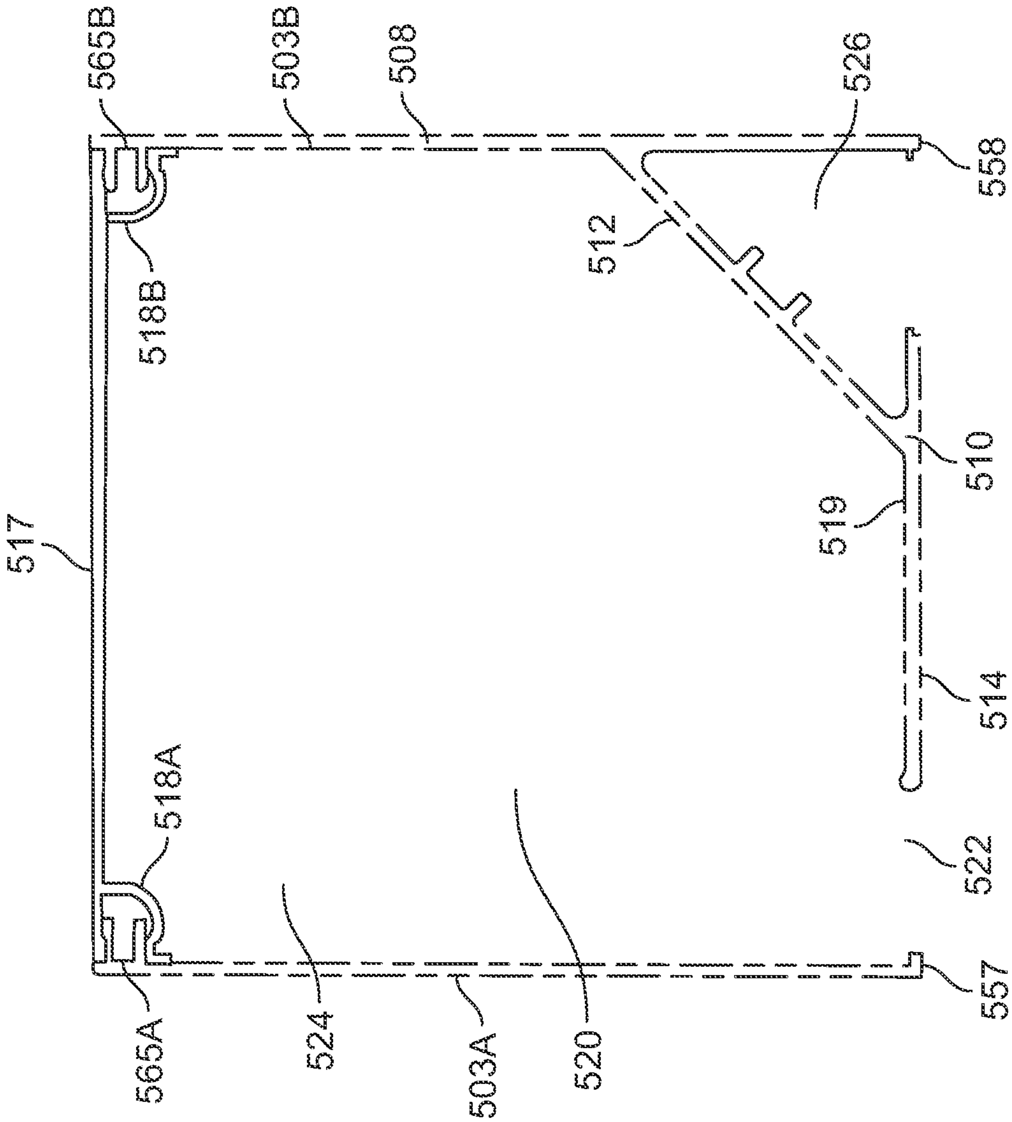
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FIG. 10G

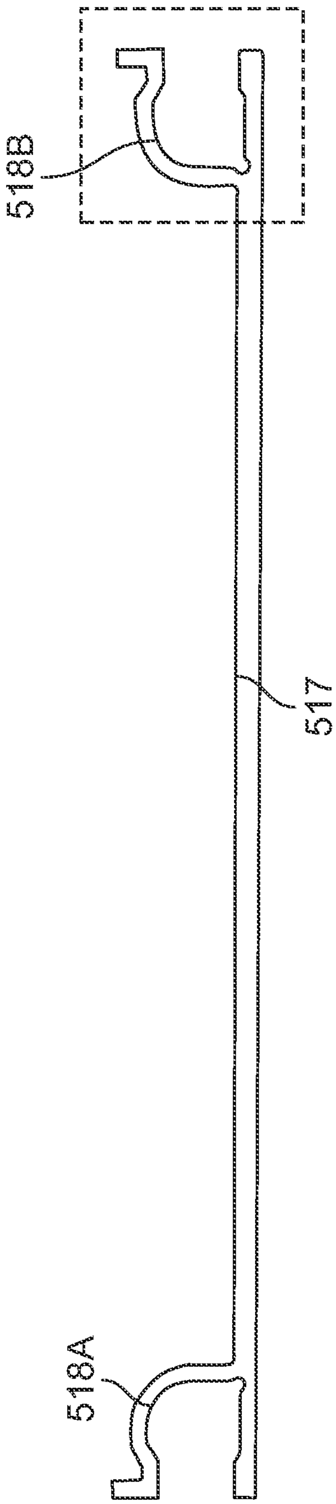

FIG. 10H

MULTIFUNCTIONAL DESIGN HOUSING COMPONENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/039,214, filed Jun. 15, 2020, and to U.S. Provisional Application No. 63/065,556, filed Aug. 14, 2020, each of which the entire contents are incorporated herein by reference.

FIELD OF THE INVENTION

The disclosure relates generally to building materials and, more particularly, to building components for outdoor structures.

BACKGROUND

Outdoor structures such as pergolas and gazebos provide an area for the enjoyment of an outdoor space. These outdoor structures can be made of wood and include vertical posts that support cross beams. The outdoor structures may be used to provide full or partial shade from the sun or rain protection, thereby allowing for a more enjoyable outdoor experience, such as for dining or recreation purposes.

SUMMARY

In some embodiments, a multifunctional design housing includes an outer case including a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge. The multifunctional design housing further includes a front opening between a first side wall front edge and a second side wall front edge, where the front opening provides access to an interior volume within the outer case. The multifunctional design housing also includes a conduit control arm, where the conduit control arm includes a support element spaced forward from the back wall and coupled to the outer case by a spacing element, and where a back surface of the conduit control arm is in contact with the interior volume.

In some embodiments, a multifunctional design housing includes an outer case including a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge. The multifunctional design housing further includes a first front opening and a second front opening between a first side wall front edge and a second side wall front edge, where the first front opening provides access to a first interior volume within the outer case and the second front opening provides access to a second interior volume within the outer case. The multifunctional design housing also includes a conduit control arm, where the conduit control arm includes a support element spaced forward from the back wall and coupled to the outer case by a spacing element. Further, a back surface of the conduit control arm is in contact with the first interior volume and the second interior volume.

In some embodiments, an ornamental structure includes at least one multifunctional design housing. The multifunctional design housing includes an outer case including a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge. The multifunctional design housing further includes a front opening between a first side wall front edge and a second side wall front edge, where the front opening provides access to an interior volume within the outer case. The multifunctional design housing also includes a conduit control arm, where the conduit control arm includes a support element spaced forward from the back wall and coupled to the outer case by a spacing element, and where a back surface of the conduit control arm is in contact with the interior volume. The ornamental structure further includes lights disposed with the interior volume, where there is an unobstructed path between the lights and the front opening.

BRIEF DESCRIPTION OF THE DRAWINGS

The features and advantages of the present disclosures will be more fully disclosed in, or rendered obvious by the following detailed descriptions of example embodiments. The detailed descriptions of the example embodiments are to be considered together with the accompanying drawings wherein like numbers refer to like parts and further wherein:

FIG. **1** illustrates an ornamental structure with multifunctional design housing components in accordance with some embodiments;

FIG. **2**A illustrates a track component of a multifunctional design housing in accordance with some embodiments;

FIG. **2**B illustrates another view of the track component of FIG. **2**A in accordance with some embodiments;

FIG. **3**A illustrates a track component of a multifunctional design housing in accordance with some embodiments;

FIG. **3**B illustrates another view of the track component of FIG. **3**A in accordance with some embodiments;

FIG. **4**A illustrates a track component of a multifunctional design housing in accordance with some embodiments;

FIG. **4**B illustrates another view of the track component of FIG. **4**A in accordance with some embodiments;

FIG. **5**A illustrates a track component of a multifunctional design housing in accordance with some embodiments;

FIG. **5**B illustrates another view of the track component of FIG. **5**A in accordance with some embodiments;

FIG. **5**C illustrates the track component of FIG. **5**A coupled to another track component in accordance with some embodiments;

FIG. **6**A illustrates a diagram of portions of the track component of FIGS. **4**A and **4**B in accordance with some embodiments;

FIG. **6**B illustrates another diagram of portions of the track component of FIGS. **4**A and **4**B in accordance with some embodiments;

FIG. **7** illustrates a cover component of a multifunctional design housing in accordance with some embodiments;

FIG. **8**A illustrates a diagram of portions of the track component of FIGS. **3**A and **3**B in accordance with some embodiments;

FIG. **8**B illustrates another diagram of portions of the track component of FIGS. **3**A and **3**B in accordance with some embodiments in accordance with some embodiments;

FIG. **9** illustrates a diagram of portions the track component of FIGS. **2**A and **2**B in accordance with some embodiments;

FIG. **10**A illustrates a diagram of portions the track component of FIGS. **5**A, **5**B, and **5**C in accordance with some embodiments;

FIG. **10**B illustrates another diagram of portions of the track component of FIGS. **5**A, **5**B, and **5**C in accordance with some embodiments;

FIG. **10**C illustrates yet another diagram of portions of the track component of FIGS. **5**A, **5**B, and **5**C in accordance with some embodiments;

FIG. **10**D illustrates a diagram of a male connector component of the track component of FIGS. **5**A, **5**B, and **5**C in accordance with some embodiments;

FIG. **10**E illustrates a diagram of a side wall that includes the male connector component of FIG. **10**D in accordance with some embodiments;

FIG. **10**F illustrates a diagram of a female connector component of the track component of FIGS. **5**A, **5**B, and **5**C in accordance with some embodiments;

FIG. **10**G illustrates the male connector component of FIG. **10**D coupled to the female connector component of FIG. **10**F in accordance with some embodiments; and

FIG. **10**H illustrates a diagram of a back wall that includes the female connector of FIG. **10**F in accordance with some embodiments.

DETAILED DESCRIPTION

The description of the preferred embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description of these disclosures. In this description, relative terms such as "horizontal," "vertical," "up," "down," "top," "bottom," as well as derivatives thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including "inwardly" versus "outwardly," "longitudinal" versus "lateral" and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as "connected" and "interconnected," refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both moveable or rigid attachments or relationships, unless expressly described otherwise, and includes terms such as "directly" coupled, secured, etc. The term "operatively coupled" is such an attachment, coupling, or connection that allows the pertinent structures to operate as intended by virtue of that relationship.

Turning to the drawings, FIG. **1** shows an example ornamental structure **100** located adjacent a house **158**. Ornamental structure **100** may be a pergola, for example, that allows for the enjoyment of an outdoor space **155**. The ornamental structure **100** can include a front beam **110**, a middle beam **112**, and a back beam **114** each positioned between a first side beam **116** and a second side beam **118**. The ornamental structure **100** further includes a plurality of louvers **108**A positioned between front beam **110** and middle beam **112**, and a plurality of louvers **108**B positioned between middle beam **112** and back beam **114**. Ornamental structure **100** also includes a first post **120**A and a second post **120**B. Each of first post **120**A and second post **120**B are coupled to front beam **110**. For example, each of first post **120**A and second post **120**B may be secured to front beam **110** with nails, glue, brackets, or any other suitable securing mechanism. Each of front beam **110**, middle beam **112**, back beam **114**, first post **120**A and second post **120**B may be manufactured out of wood, plastic, plastic composite, metal (e.g., aluminum), or any other suitable material. Louvers **108**A, **108**B may be manufactured out of wood, plastic, plastic composite, metal (e.g., aluminum), or any other suitable material.

In the example of FIG. **1**, a portion of the ornamental structure **100** is supported by the house **158**. However, the portion of the ornamental structure **100** supported by the house could also be supported at the corners by additional posts **120**C, **120**D.

Ornamental structure **100** further includes various multifunctional design housings that can hold components such as wiring and motorized screens. The wiring may provide power to various components including lighting (e.g., LED lighting), and to power electrical devices, such as motorized screens, motorized louvres, ceiling fans, etc. that are attached or used in connection with the ornamental structure **100**. In this example, the multifunctional design housings include a first hood track **104**A and a second hood track **104**B. Each of first hood track **104**A and second hood track **104**B are secured to front beam **110**. For example, the first hood track **104**A and the second hood track **104**B may be secured to front beam **110** with nails, screws, glues, brackets, or any other suitable securing mechanisms.

Although illustrated as a single first hood track **104**A, in some examples, first hood track **104**A includes multiple first hood tracks **104**A coupled to each other longitudinally to span longer distances. In addition, each of the first hood track **104**A and second hood track **104**B may include LED strip lighting **105**A, **105**B, respectively.

Multifunction design housings also include a first column track **106**A and a second column track **106**B. First column track **106**A is secured to second post **120**B, and second column track **106**B is secured to first post **120**A. Although illustrated as a single first column track **106**A, in some examples, first column track **106**A includes multiple column tracks **106**A coupled to each other (e.g., with clips, brackets, or attached to post **120**B). Similarly, second column track **106**B may be a single second column track **106**B or, in some examples, may include multiple second column track **106**B coupled to each other.

In addition, first column track **106**A includes LED strip lighting **107**A, **107**B. Similarly, second column track **106**B includes LED strip lighting **107**C, **107**D.

The multifunction design housings, including first hood track **104**A, second hood track **104**B, first column track **106**A, and second column track **106**B, may each be manufactured out of wood, plastic, plastic composite, metal (e.g., aluminum), or any other suitable material.

FIGS. **2**A and **2**B illustrate a track component **200** that includes an outer case **202** with a back wall **204**, a first side wall **206**, and a second side wall **208** opposite the first side wall **206**. First side wall **206** extends forward from a first back wall edge **207**, and second side wall **208** extends forward from a second back wall edge **209**. Further, track component **200** includes a front opening **222** between a first side wall front edge **215** and a second side wall front edge **217**. The front opening **222** provides access to an interior volume **220** within the outer case **202**.

Track component **200** also includes a conduit control arm **210** with a support element **214** spaced forward from the back wall **204** and coupled to the outer case **202** by a spacing element **212**, where a front surface **218** of the support element **214** is in contact with the interior volume **220**. In some examples, a lateral cross-section of the conduit control arm **210** is generally L-shaped. In some examples, support element **214** includes one or more lights **230**, such as light-emitting diodes (LEDs). In some embodiments, the one or more lights **230** can be secured along the front surface **218** of the support element **214**.

Spacing element **212** extends from back wall **204**, thereby allowing room for components, such as wiring, to be supported by or run along a back surface **219** of the of the support element **214**. For example, as illustrated in FIG. 2B, one or more cables **280** may be supported by or run along the back surface **219** of the of the support element **214**. In this example, the back surface **219** of the support element **214** is in contact with the interior volume **220**.

In some examples, track component **200** includes a uniform lateral cross-section (e.g., from first side wall **206** to second side wall **208** across front opening **222**). In some examples, the first back wall edge **207** and the second back wall **209** edge extend longitudinally (e.g., on either side of conduit control arm **210**).

In this example, front opening **222** includes an elongated slot **224** defined by a first front opening edge **225** and a second front opening edge **227**. In some examples, the support element **214** of the conduit control arm **210** is centered behind the elongated slot **224**.

In some examples, track component **200** includes a first front opening covering **260** adapted to be removably coupled to the first and second front opening edges **225**, **227**. FIG. **2**A illustrates first front opening covering **260** removed from the first and second front opening edges **225**, **227**, while FIG. 2B illustrates first front opening covering **260** coupled to the first and second front opening edges **225**, **227**. The conduit control arm **210** is spaced apart from the front opening covering **260** when the first front opening covering **260** is removably coupled to the first and second front opening edges **225**, **227**. To accommodate the coupling, front opening covering **260** may include a first channel **262** and a second channel **264**. First channel **262** and second channel **264** are adapted to be removably coupled to the first and second front opening edges **225**, **227**. For example, each of the first and second front opening edges **225**, **227** may "snap" or slide into first channel **262** and second channel **264**, respectively, to secure the first front opening covering **260** to outer case **202**. In some examples, first front opening covering **260** is translucent or transparent in order to obtain a desired lighting effect.

Track **200** may be employed as a hood track, such as first hood track **104**A or second hood track **104**B, or as a column track, such as first column track **106**A or second column track **106**B, of an ornamental structure, such as ornamental structure **100**.

FIG. **9** illustrates lateral cross-sectional diagram of track component **200**. The arrows indicate distances that may be determined prior to manufacturing.

FIGS. **3**A and **3**B illustrate a track component **300** includes an outer case **303** with a back wall **317**, a first side wall **350**, and a second side wall **352** opposite the first side wall **350**. First side wall **350** extends forward from a first back wall edge **354**, and second side wall **352** extends forward from a second back wall edge **356**. Further, track component **300** includes a first front opening **322**A and a second front opening **322**B between a first side wall front edge **357** and a second side wall front edge **358**. The first front opening **322**A provides access to a first interior volume **320**A within the outer case **303**, and the second front opening **322**B provides access to a second interior volume **320**B within the outer case **303**.

Track component **300** also includes a conduit control arm **310** with a support element **314** spaced forward from the back wall **317** and coupled to the outer case **303** by a spacing element **312**, where a back surface **319** of the conduit control arm **310** is in contact with the first interior volume **320**A and the second interior volume **320**B. First interior volume **320**A and second interior volume **320**B are separated by spacing element **312**. In some examples, the first interior volume **320**A and the second interior volume **320**B include a similar amount of space (e.g., each may include near 50% of their total volume). In some examples, heights of the first side wall **350**, the second side wall **352**, and the conduit control arm **310** are the same. In this example, a front surface **318** of the conduit control arm **310** is on an exterior of the track component **300**.

In some examples, spacing element **314** extends from the back wall **317** of the outer case **303** to an intermediate portion of the support element **314**. In some examples, a lateral cross-section of the conduit control arm **310** is generally T-shaped. Support element **314** includes a first support element longitudinal edge **340** and a second support element longitudinal edge **342** opposite the first support element longitudinal edge **340**. In some examples, at least one side of the support element **314** includes a conduit retention ridge extending rearwardly. In this example, support element **314** includes a first conduit retention ridge **362** and a second conduit retention ridge **364** each extending rearwardly. As illustrated in FIG. **3**B, one or more sets of cables **280** may be supported by or run along a back surface **319** of the support element **314** on either side of spacing element **314**.

In this example, first front opening **322**A includes a first slot **301** and second front opening **322**B includes a second slot **305**, each of the first slot **301** and second slot **305** located on opposite sides of the spacing element **314**. The first slot **301** is defined by a first slot inner edge **302** and a first slot outer edge **304**, and the second slot **305** is defined by a second slot inner edge **306** and a second slot outer edge **308**. As illustrated, in some examples the opposing first and second longitudinal edges **340**, **342** of the support element **314** include the first slot inner edge **302** and the second slot inner edge **306**, respectively. In some examples, the first side wall **350** and the second side wall **352** include the first slot outer edge **304** and the second slot outer edge **308**, respectively.

In some examples, the back wall **317** includes at least one channel aligned with either the first slot **301** or the second slot **305**. In this example, the back wall **317** includes a first channel **370** aligned with the first slot **301**, and a second channel **372** aligned with the second slot **305**.

In some examples, track component **300** includes a first slot covering **390** adapted to be removably coupled to the first slot inner edge **302** and the first slot outer edge **304**. For example, the first slot covering **390** may be slidably coupled, snapably coupled, or screwed to each of the first slot inner edge **302** and the first slot outer edge **304**. In some examples, track component includes a second slot covering **380** adapted to be removably coupled to the second slot inner edge **306** and the second slot outer edge **308**. For example, the second slot covering **380** may be slidably coupled, snapably coupled, or screwed to each of the second slot inner edge **306** and the second slot outer edge **308**.

In some examples, at least one of the first slot covering **390** and the second slot covering **380** are translucent or transparent. For example, each of the first slot covering **390** and the second slot covering **380** may be translucent or transparent. While FIG. **3**A illustrates first slot covering **390** and second slot covering **380** removed the support element **314**, FIG. **3**B illustrates first slot covering **390** removably coupled to the first slot inner edge **302** and the first slot outer edge **304**, and second slot covering **380** removably coupled to the second slot inner edge **306** and the second slot outer edge **308**.

Track **300** may be employed as a hood track, such as first hood track **104**A or second hood track **104**B, or as a column track, such as first column track **106**A or second column track **106**B, of an ornamental structure, such as ornamental structure **100**.

FIGS. **8**A and **8**B illustrate diagrams of track component **300**. Referring to FIG. **8**A, track component **300** includes a first back wall channel **832** within first interior volume **320**A. First back wall channel **832** is defined by a first back wall channel first wall **833** and a first back wall channel second wall **830**. First back wall channel **832** may run longitudinally along first channel **370**. Track component **300** also includes a second back wall channel **842** within second interior volume **320**B. Second back wall channel **842** is defined by a second back wall channel first wall **843** and a second back wall channel second wall **840**. Second back wall channel **842** may run longitudinally along second channel **372**. The arrows in FIG. **8**B indicate distances that may be determined prior to manufacturing.

FIGS. **4**A and **4**B illustrate a track component **400** that includes an outer case **403** with a back wall **417**, a first side wall **450**, and a second side wall **452** opposite the first side wall **450**. First side wall **450** extends forward from a first back wall edge **454**, and second side wall **452** extends forward from a second back wall edge **456**. Further, track component **400** includes a first front opening **422**A and a second front opening **422**B between a first side wall front edge **457** and a second side wall front edge **458**. The first front opening **422**A provides access to a first interior volume **420**A within the outer case **403**, and the second front opening **422**B provides access to a second interior volume **420**B within the outer case **403**.

Track component **400** also includes a conduit control arm **410** with a support element **414** spaced forward from the back wall **417** and coupled to the outer case **403** by a spacing element **412**, where a back surface **419** of the conduit control arm **410** is in contact with the first interior volume **420**A and the second interior volume **420**B. First interior volume **420**A and the second interior volume **420**B are separated by spacing element **412**. In some examples, the first interior volume **420**A and the second interior volume **420**B include a similar amount of space (e.g., each may include near 50% of their total volume). In some examples, heights of the first side wall **450**, the second side wall **452**, and the conduit control arm **410** are the same. In this example, a front surface **418** of the conduit control arm **410** is on an exterior of the track component **400**.

In some examples, spacing element **414** extends from the back wall **417** of the outer case **403** to an intermediate portion of the support element **414**. In some examples, the conduit control arm **410** is generally T-shaped. Support element **414** includes a first support element longitudinal edge **440** and a second support element longitudinal edge **442** opposite the first support element longitudinal edge **440**. In some examples, at least one side of the support element **414** includes a conduit retention ridge extending rearwardly. In this example, support element **414** includes a first conduit retention ridge **462** and a second conduit retention ridge **464** each extending rearwardly. As illustrated in FIG. **4**B, one or more sets of cables **280** may be run along a back surface **419** of the support element **414** on either side of spacing element **414**.

In this example, first front opening **422**A includes a first slot **401** and second front opening **422**B includes a second slot **405**, each of the first slot **401** and second slot **405** located on opposite sides of the spacing element **414**. The first slot **401** is defined by a first slot inner edge **402** and a first slot outer edge **404**, and the second slot **405** is defined by a second slot inner edge **406** and a second slot outer edge **408**. As illustrated, in some examples the opposing first and second longitudinal edges **440**, **442** of the support element **414** include the first slot inner edge **402** and the second slot inner edge **406**, respectively. In some examples, the first side wall **450** and the second side wall **452** include the first slot outer edge **404** and the second slot outer edge **408**, respectively.

In some examples, the back wall **417** includes at least one channel aligned with either the first slot **401** or the second slot **405**. In this example, the back wall **417** includes a first channel **470** aligned with the first slot **401**, and a second channel **472** aligned with the second slot **405**.

In some examples, track component **400** includes a first slot covering **490** adapted to be removably coupled to the first slot inner edge **402** and the first slot outer edge **404**. For example, the first slot covering **490** may be slidably coupled, snapably coupled, or screwed to each of the first slot inner edge **402** and the first slot outer edge **404**. In some examples, track component includes a second slot covering **480** adapted to be removably coupled to the second slot inner edge **406** and the second slot outer edge **408**. For example, the second slot covering **480** may include a first edge channel **482** and a second edge channel **484** that can be slidably coupled, snapably coupled, or screwed to each of the second slot inner edge **406** and the second slot outer edge **408**, respectively.

In some examples, at least one of the first slot covering **490** and the second slot covering **480** are translucent or transparent. For example, each of the first slot covering **490** and the second slot covering **480** may be translucent or transparent. While FIG. **4**A illustrates first slot covering **490** and second slot covering **480** removed the support element **414**, FIG. **4**B illustrates first slot covering **490** removably coupled to the first slot inner edge **402** and the first slot outer edge **404**, and second slot covering **480** removably coupled to the second slot inner edge **406** and the second slot outer edge **408**.

Track **400** may be employed as a hood track, such as first hood track **104**A or second hood track **104**B, or as a column track, such as first column track **106**A or second column track **106**B, of an ornamental structure, such as ornamental structure **100**. Track component **400**

FIGS. **6**A and **6**B illustrate diagrams of track component **400**. Referring to FIG. **6**A, track component **400** includes a first back wall channel **632** within first interior volume **420**A. First back wall channel **632** is defined by a first back wall channel first wall **633** and a first back wall channel second wall **630**. First back wall channel **632** may run longitudinally along first channel **470**. Track component **400** also includes a second back wall channel **642** within second interior volume **420**B. Second back wall channel **642** is defined by a second back wall channel first wall **643** and a second back wall channel second wall **640**. Second back wall channel **642** may run longitudinally along second channel **472**. The arrows in FIG. **9**B indicate distances that may be determined prior to manufacturing.

FIGS. **5**A, **5**B, and **5**C illustrate a hood component **500** that may be employed as a hood track, such as first hood track **104**A or second hood track **104**B, of an ornamental structure, such as ornamental structure **100**. Hood component **500** includes an outer case **503** with a back wall **517**, a first side wall **506**, and a second side wall **508** opposite the first side wall **506**. First side wall **506** extends forward from a first back wall edge **554**, and second side wall **508** extends forward from a second back wall edge **556**. First side wall **506** includes a first end tab **565**A, and second side wall **508** includes a second end tab **565**B. First end tab **565**A and second end tab **565**B may extend longitudinally along the length of first side wall **506** and second side wall **508**, respectively. First end tab **565**A can be slidably coupled, snapably coupled, or suitable coupled to a first back wall channel **518**A. Similarly, second end tab **565**B can be slidably coupled, snapably coupled, or otherwise suitably coupled to a second back wall channel **518**B. In some examples, there is a friction fit between first end tab **565**A and first back wall channel **518**A, and between second end tab **565**B and second back wall channel **518**B.

Further, track component **500** includes a first front opening **522**A and a second front opening **522**B between a first side wall front edge **557** and a second side wall front edge **558**. The first front opening **522**A provides access to a first interior volume **520**A within the outer case **503** and the second front opening **522**B provides access to a second interior volume **520**B. Track component **500** also includes a conduit control arm **510** with a spacing element **512** that extends from the second side wall **508** to an intermediate portion of a support element **514**. A back surface **519** of the conduit control arm **510** is in contact with each of the first interior volume **520**A and the second interior volume **520**B.

First interior volume **520**A and second interior volume **520**B are separated by spacing element **512**. In some examples, a space of the second interior volume **520**B is less than 40% of the space of the total of the first interior volume **520**A and the second interior volume **520**B. In some examples, maximum distances from the back wall **517** to an end of the first side wall **506**, an end of the second side wall **508**, and the support element **514** are the same. In some examples, a front surface **516** of the conduit control arm **510** is on an exterior of the track component **500**. A space element channel **572** extends along a surface of the space element **512** and is in contact with the second interior volume **520**B.

First front opening **522**A includes a first slot **511** and second front opening **522**B includes a second slot **513**, each of first front opening **522**A and second front opening **522**B located on opposite sides of the spacing element **512**. The first slot **511** is defined by a first slot inner edge **537** and a first slot outer edge **535**, and the second slot **513** is defined by a second slot inner edge **527** and a second slot outer edge **525**.

In some examples, hood component **500** includes a slot covering **590** adapted to be removably coupled to second slot inner edge **527** and second slot outer edge **525**. For example, the slot covering **590** may include a first edge channel **582** and a second edge channel **584** that can be slidably coupled, snapably coupled, or screwed to each of the second slot inner edge **527** and the second slot outer edge **525**, respectively. While FIG. **5**A illustrates slot covering **590** removed the support element **514**, FIG. **5**B illustrates slot covering **590** removably coupled to the second slot inner edge **527** and the second slot outer edge **525**.

Hood component **500** further includes a screen roll **591** that may extend through the length of the first interior volume **520**A of one or more hood components **500**. In this example, the screen roll **591** is part of a retractable screen assembly **592** that includes a retractable screen **593**, a weighting member **594** to assist during the lowering of the retractable screen **593**, and bristles **596** that may contact the ground when the retractable screen **593** is lowered. The screen roll **591** may be electrically rotated to lower, or raise, the retractable screen **593**. In some examples, retractable screen assembly **592** includes a guide rod that extends through first front opening **522**A. For example, the guide rod may support and allow the screen roll to rotate. The guide rod may extend through front openings of other components, such as through first front openings **522**A of other hood components **500**.

FIG. **5**C illustrates hood component **500** coupled to a track component **400** along interface plane **561**. Hood component **500** may be secured to the track component **300**, **400** with nails, screws, glue, brackets, via soldering, or any other suitable securing mechanism. In some examples, hood component **500** may be attached to adjacent portions of ornamental structure **100**. In this example, retractable screen **593** can raise and lower through first interior volume **420**A of track component **400**. As such, in this example, track component **400** does not include first slot covering **490**.

FIGS. **10**A, **10**B, **10**C, and **10**G illustrate portions of hood component **500**. FIG. **10**D illustrates a diagram of an end tab **565**, such as first end tab **565**A or second end tab **565**B. End tab **565** includes a channel **1001** defined by a first wall **1002**, a second wall **1003**, and a third wall **1004**. FIG. **10**E illustrates a diagram of first side wall **506** with an end tab **565**. FIGS. **10**F illustrates a diagram of back wall **517** including a back wall channel **518**, such as first back wall channel **518**A and second back wall channel **518**B. Back wall channel **518** is defined by a first wall **1010** that curves radially inward and includes a first wall end **1016**, and a second wall **1012** that includes a second wall end **1018**. End tab **565** may couple to back wall channel. For example, end tab **565** may "snap" into back wall channel **518**. In some examples, a friction fit is formed when end tab **565** is inserted into back wall channel **518**. A maximum distance between first wall **1010** and second wall **1012** may, in some examples, be greater than a maximum distance between first wall end **1016** and second wall end **1018**. FIG. **10**H illustrates a diagram of back wall **517** with first back wall channel **518**A and second back wall channel **518**B.

FIG. **7** illustrates a diagram of an exemplary slot covering **700**, such as first slot covering **490** and second slot covering **480** of FIGS. **4**A and **4**B, first slot covering **390** and second slot covering **380** of FIGS. **3**A and **3**B, or slot covering **590** of FIGS. **5**A, **5**B, and **5**C. Slot covering **700** includes a first end **783** with a first edge channel **784**, and a second end **781** with a second edge channel **782**. The arrows indicate distances that may be determined prior to manufacturing. Slot covering **700** may be manufactured such that it will couple to track component **200**, track component **300**, track component **400**, or hood component **500**.

In some examples, a multifunctional design housing includes an outer case comprising a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge. The multifunctional design housing also includes a front opening between a first side wall front edge and a second side wall front edge, where the front opening provides access to an interior volume within the outer case. Further, the multifunctional design housing includes a conduit control arm with a support element spaced forward from the back wall and coupled to the outer case by a spacing element. A back surface of the conduit control arm is in contact with the interior volume.

In some examples, the multifunctional design housing has a uniform lateral cross-section. In some examples, the first back wall edge and the second back wall edge extend longitudinally.

In some examples, the spacing element extends from the back wall. In some examples, the front opening comprises an elongated slot defined by a first front opening edge and a second front opening edge. The multifunctional design housing may further include a first front opening covering adapted to be removably coupled to the first and second front opening edges. In some examples, the conduit control arm is spaced apart from the front opening covering, where the first front opening covering is removably coupled to the first and second front opening edges. In some examples, the first front opening covering is translucent or transparent. In some examples, the conduit control arm is generally L-shaped. In some examples, the conduit control arm is centered behind the elongated slot. In some examples, a front surface of the conduit control arm is in contact with the interior volume.

In some examples, the spacing element extends from the outer case to an intermediate portion of the support element. In some examples, the front opening includes a first slot and a second slot located on opposite sides of the spacing element. The first slot is defined by a first slot inner edge and a first slot outer edge, and the second slot is defined by a second slot inner edge and a second slot outer edge.

In some examples, the multifunctional design housing includes a first slot covering adapted to be removably coupled to the first slot inner edge and the first slot outer edge. In some examples, the multifunctional design housing includes, additionally or alternatively, a second slot covering adapted to be removably coupled to the second slot inner edge and the second slot outer edge. In some examples, the first slot covering, the second slot covering, or both are translucent or transparent.

In some examples, the opposing longitudinal edges of the support element include the first slot inner edge and the second slot inner edge. Additionally or alternatively, the first side wall and the second side wall include the first slot outer edge and the second slot outer edge respectively.

In some examples, the interior volume includes a first interior volume and a second interior volume, which are separated by the spacing element. In some examples, at least one side of the support element includes a conduit retention ridge extending rearwardly. In some examples, heights of the first side wall, the second side wall, and the conduit control arm are the same.

In some examples, a front surface of the conduit control arm is on an exterior of the multifunctional design housing. In some examples, the spacing element extends from the back wall to an intermediate portion of the support element. In some examples, the back wall comprises a first channel aligned with the first slot, a second channel aligned with the second slot, or both. In some examples, the spacing element extends from the back wall to an intermediate portion of the support element.

In some examples, the interior volume includes a first interior volume and a second interior volume, which are separated by the spacing element. In some examples, the second interior volume is less than 40% of a total of the first interior volume and the second interior volume.

In some examples, the front opening includes a first slot and a second slot located on opposite sides of the spacing element. In some examples, the first slot is defined by a first slot inner edge and a first slot outer edge, and the second slot is defined by a second slot inner edge and a second slot outer edge.

In some examples, maximum distances from the back wall are the same for the first side wall, the second side wall, and the conduit control arm. In some examples, a front surface of the conduit control arm is on an exterior of the multifunctional design housing.

In some examples, an ornamental structure includes at least one multifunctional design housing as described herein. For example, the multifunctional design housing may include an outer case comprising a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge. The multifunctional design housing also includes a front opening between a first side wall front edge and a second side wall front edge, where the front opening provides access to an interior volume within the outer case. Further, the multifunctional design housing includes a conduit control arm with a support element spaced forward from the back wall and coupled to the outer case by a spacing element. A back surface of the conduit control arm may be in contact with the interior volume.

In some examples, the ornamental structure includes lights disposed within the interior volume of the at least one multifunctional design housing, where there is an unobstructed path between the lights and the front opening.

In some examples, the ornamental structure includes a first multifunctional design housing and a second multifunctional design housing arranged parallel to one another, where a guide rod extends through front openings of the first and second multifunctional design housings.

In some examples, the ornamental structure includes a third multifunctional design housing. The third multifunctional design housing may include a front opening aligned with the front openings of the first and second multifunctional design housings. In some examples, the guide rod is part of a retractable screen anchored in the third multifunctional design housing.

The foregoing is provided for purposes of illustrating, explaining, and describing embodiments of these disclosures. Modifications and adaptations to these embodiments will be apparent to those skilled in the art and may be made without departing from the scope or spirit of these disclosures.

What is claimed is:

1. A multifunctional design housing, comprising:

an outer case comprising a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge;

a front opening between a first side wall front edge and a second side wall front edge,

wherein the front opening provides access to an interior volume within the outer case; and

a conduit control arm, said conduit control arm comprising a support element and a spacing element, the spacing element coupled to and extending from a portion of an interior surface of the back wall that is off-centered from the front opening and the support element spaced forward from the back wall and centered behind the front opening, wherein a back surface of the support element is in contact with the interior volume, and wherein a front surface of the support element includes at least one light disposed thereon.

2. The multifunctional design housing as claimed in claim **1**, wherein the multifunctional design housing has a uniform lateral cross-section.

3. The multifunctional design housing as claimed in claim **1**, wherein the first back wall edge and the second back wall edge extend longitudinally.

4. The multifunctional design housing as claimed in claim **1**, wherein:

the front opening comprises an elongated slot defined by a first front opening edge and a second front opening edge; and

the multifunctional design housing further comprises a front opening covering adapted to be removably coupled to the first and second front opening edges.

5. The multifunctional design housing as claimed in claim **4**, wherein the conduit control arm is spaced apart from the front opening covering when the front opening covering is removably coupled to the first and second front opening edges.

6. The multifunctional design housing as claimed in claim **5**, wherein the front opening covering is translucent or transparent.

7. The multifunctional design housing as claimed in claim **1**, wherein the conduit control arm is generally L-shaped.

8. The multifunctional design housing as claimed in claim **1**, wherein a front surface of the conduit control arm is in contact with the interior volume.

9. The multifunctional design housing as claimed in claim **1**, wherein the spacing element extends to an intermediate portion of the support element.

10. The multifunctional design housing as claimed in claim **1**, wherein at least one side of the support element comprises a conduit retention ridge extending rearwardly toward the interior surface of the back wall.

11. The multifunctional design housing as claimed in claim **1**, wherein a front surface of the conduit control arm is on an exterior of the multifunctional design housing.

12. The multifunctional design housing as claimed in claim **1**, wherein the interior volume comprises a first interior volume and a second interior volume separated from the first interior volume by the spacing element.

13. The multifunctional design housing as claimed in claim **12**, wherein the second interior volume is less than 40% of a total of the first interior volume and the second interior volume.

14. The multifunctional design housing as claimed in claim **1**, wherein the front opening comprises a first slot and a second slot located on opposite sides of the spacing element, wherein the first slot is defined by a first slot inner edge and a first slot outer edge, wherein the second slot is defined by a second slot inner edge and a second slot outer edge.

15. The multifunctional design housing as claimed in claim **14**, wherein:

opposing longitudinal edges of the support element comprise the first slot inner edge and the second slot inner edge; and

the first side wall and the second side wall comprise the first slot outer edge and the second slot outer edge respectively.

16. The multifunctional design housing as claimed in claim **15**, further comprising at least one of:

a first slot covering adapted to be removably coupled to the first slot inner edge and the first slot outer edge; and

a second slot covering adapted to be removably coupled to the second slot inner edge and the second slot outer edge.

17. An ornamental structure comprising:

at least one multifunctional design housing comprising:

an outer case comprising a back wall, a first side wall extending forward from a first back wall edge and a second side wall extending forward from a second back wall edge;

a front opening between a first side wall front edge and a second side wall front edge, wherein the front opening provides access to an interior volume within the outer case; and

a conduit control arm, said conduit control arm comprising a support element and a spacing element, the spacing element coupled to and extending from a portion of an interior surface of the back wall that is off-centered from the front opening and the support element spaced forward from the back wall and centered behind the front opening, wherein a back surface of the support element is in contact with the interior volume; and

at least one light disposed within the interior volume and disposed on a front surface of the support element, wherein there is an unobstructed path between the lights and the front opening.

18. The ornamental structure as claimed in claim **17**, wherein the at least one multifunctional design housing comprises a first multifunctional design housing and a second multifunctional design housing arranged parallel to one another, wherein a guide rod extends through front openings of the first and second multifunctional design housings.

19. The ornamental structure as claimed in claim **18**, further comprising a third multifunctional design housing, wherein the third multifunctional design housing comprises a front opening aligned with the front openings of the first and second multifunctional design housings, and where the guide rod is part of a retractable screen anchored in the third multifunctional design housing.

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